The Rank–Value Gap: Data Poisoning Attacks on Locally Private Tail Quantile Estimation

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Abstract*—Tail quantiles of user data, such as p99 latencies or transaction amounts, are thresholds that operational systems act on, and local differential privacy (LDP) is a principled way to collect them from untrusted devices. We show why LDP tail quantiles are cheap to poison. Above the q-th quantile, the cumulative distribution function is confined to $[q, 1]$ (*tail saturation*), so every threshold decision in the upper domain has a margin of at most $1 - q$, however far the threshold lies from the true quantile: moving a tail quantile costs *rank*, not *value*. In our proposed attack, CLIFF, fake users answer every threshold query as noise-free holders of the attacker's target $m^\star$, and the accuracy guarantee of threshold-oracle protocols pins the reported quantile there. As the protocol's error vanishes, any fake-user fraction above $(1-q)(1-e^{-\varepsilon})$ places the q-th quantile anywhere above its true value; for targets of rank near 1, this budget is asymptotically optimal among monotone time-invariant strategies, and LDP debiasing makes it, asymptotically, at least $1/(1-e^{-\varepsilon})$ times smaller than for input poisoning. At finite n, the error delays the advantage over non-private contamination: on a real transaction dataset at $\varepsilon = 1$, 1% fake users, as many as a non-private p99 needs, place the p99 within 10% of ten times its value in 88–95% of runs against our implementations of the two state-of-the-art protocols with 10^6 users, and 0.74% suffice in about 90% of runs against noisy binary search with 10^7. We also design CTQ, a defense that provably caps inflation with high probability, given a bound on the fake fraction and a certified protocol error, which we derive for noisy binary search. We prove that its rank shift is necessary, up to the protocol's error, for any estimator that collects one threshold answer per user. Finally, on noisy binary search, an audit derived from tail saturation detects CLIFF only with large audit cohorts, and a lowered step evades it.

***Index Terms*—local differential privacy, quantile estimation, data poisoning, robustness, threshold queries**

I. Introduction

Systems act on tail quantiles. Service-level objectives (SLOs) are written on p95 and p99 latencies [1], payment systems route transactions above a high percentile of the amount distribution to manual review, telemetry pipelines flag devices whose readings exceed a high percentile, and differentially private federated learning adapts its clipping norm to a quantile of client update norms [2]. When these statistics come from user devices, the collector usually cannot be trusted with raw values. *Local differential privacy* (LDP) [3]–[5] addresses this: each device randomizes its own report, so that the report reveals little about its value, as quantified by a privacy parameter ε, and LDP has been deployed at scale by Google [6], Apple [7], and Microsoft [8]. A dedicated line of work estimates quantiles under LDP, using hierarchical range-query structures [9], [10], online stochastic approximation with self-normalized inference [11], and adaptive noisy binary search with optimal sample complexity [12].

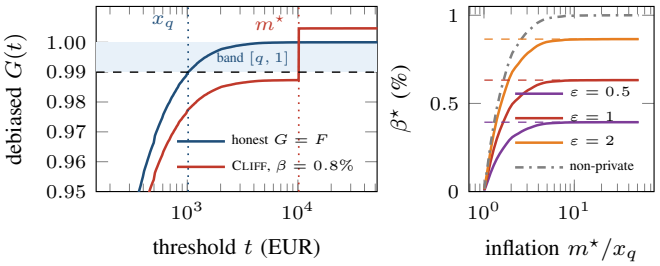


Fig. 1. Tail saturation on CreditCard ($q = 0.99$). *Left* ($\varepsilon = 1$): above the true p99 x_q, the honest CDF F lies in $[q, 1]$. With 0.8% fake users, CLIFF pushes the debiased estimate G below q on $[x_q, m^\star)$ and above q from $m^\star = 10x_q$ on, so a protocol with error below these margins reports $m^\star$. *Right*: the asymptotic budget $\beta_0^\star$ to pin the p99 at $m^\star$ (Theorem 1) saturates at $(1-q)(1-e^{-\varepsilon})$ (dashed) however large $m^\star/x_q$, below the non-private breakdown point.

The same openness that makes LDP attractive makes it fragile. Anyone can register fake devices, and because the collector cannot see raw values, fake reports cannot be told apart from honest ones. A growing literature shows how such fake users can skew LDP frequencies, means, numerical distributions, range queries, and other statistics (§II). Quantiles, the statistics that most directly set thresholds, have been studied only for robust median estimation under contamination [13], not as a targeted attack goal (untargeted attacks move them only as a side effect, §II). Attacks on means and frequencies shift an estimate by injecting mass, and their cost grows with how far it must move; for a quantile, in contrast, it is set by rank, and LDP debiasing lowers it.

Consider a quantile level q close to 1, e.g., $q = 0.99$ for the p99. Every LDP quantile protocol we know of decides, for some threshold t, on which side of q the fraction of values at most t lies. Above the true quantile x_q, the cumulative distribution function (CDF) F of the honest values lives in the band $[q, 1]$ (Fig. 1, left). Every threshold decision in the upper domain therefore has a margin of at most $1-q$, whether the threshold sits 10% or 100× above x_q. Up to the protocol's error, the cost of moving the q-th quantile to a value $m^\star$ is governed by the *rank* $F(m^\star - 1) - q \le 1 - q$, while the *value* displacement $m^\star - x_q$ is free. This is the *rank–value gap*. It is invisible to the accuracy notions used to design LDP quantile protocols, which bound the rank error of the output. Aamand et al. [12] note that for quantile levels within the rank-error

tolerance of 0 or 1, simply returning an endpoint of the domain already meets the guarantee. Moreover, Cormode et al. [9] found that large value errors concentrate in sparse tails while quantile (rank) errors stay flat, and read this as benign. We show, in contrast, that this gap is an attack surface.

Exploiting tail saturation, however, requires two things: pushing the response curve below q on $[x_q, m^\star)$ as cheaply as possible, and making the protocol stop at exactly $m^\star$ rather than somewhere in a flat tail. Our attack, CLIFF, achieves both with one rule: every fake user answers every threshold query as a noise-free user holding $m^\star$, saying "no" below $m^\star$ and "yes" from $m^\star$ on. After the collector's debiasing, each "no" counts as $-1/(e^\varepsilon - 1)$, the largest downward influence a report can have under randomized response. The two sides form a cliff in the estimated CDF at $m^\star$. Because quantile protocols return the crossing point of a monotone response curve, their own correctness guarantee thus hands the attacker the target.

Contributions. Our contributions are shown below.

- We give the first analysis of targeted poisoning of threshold-query LDP quantile protocols and trace their fragility to tail saturation, the classical $1 - q$ breakdown of upper quantiles seen through threshold queries, which LDP debiasing amplifies (§IV).
- We prove that CLIFF pins the reported q-th quantile at any target $m^\star$ once the fake-user fraction exceeds a threshold that depends only on the target's ranks, q, ε, and the protocol's error, and that, as this error vanishes, never exceeds $(1 - q)(1 - e^{-\varepsilon})$. Moreover, this budget is asymptotically optimal among time-invariant fake-user strategies that keep the response curve monotone; as the error vanishes, it is at least $1/(1 - e^{-\varepsilon})$ times smaller than for input poisoning or poisoning without privacy, and it is up to e^ε times smaller for inflation than for deflation by the same rank. These asymptotic statements need the protocol's error to be small against $1 - q$; for noisy binary search, whose error we certify, this holds from about 3×10^6 users at $\varepsilon = 1$.
- We propose CTQ, which runs any accurate protocol at a shifted level and certifies, with probability at least $1 - \delta$, that the output is not inflated above the true quantile; we prove that, for any estimator that collects one randomized threshold answer per user, its shift is necessary up to the protocol's estimation error, whatever binary randomizer is used. Tail saturation also yields a jump audit that detects CLIFF's step given enough users, which a lowered step evades. Furthermore, precision matters: a change monitor that flags an untargeted push in every run misses CLIFF ramped up in small steps.

II. Background and Related Work

Local differential privacy. A randomized algorithm $\mathcal{M}$ satisfies ε-LDP if $\Pr[\mathcal{M}(x) = y] \le e^\varepsilon \Pr[\mathcal{M}(x') = y]$ for all inputs x, x' and outputs y [4], [5], [14]. Binary randomized response [3], $\mathrm{RR}_\varepsilon(b)$, reports a bit b truthfully with probability $e^\varepsilon/(1 + e^\varepsilon)$ and flips it otherwise; it underlies frequency oracles [15], [16] and the deployed systems above.

Quantile and range estimation under LDP. Non-adaptive approaches estimate a full CDF: hierarchical histograms and wavelets [9], [17] with consistency post-processing [18], [19], adaptive decompositions [10], [20], [21], multi-dimensional analytics [22], and numerical distribution estimators [23], [24]. In contrast, adaptive approaches query *thresholds*: each user reveals, through randomized response, whether its value lies below a threshold chosen by the collector. In particular, noisy binary search [25] and the Bayesian screening search of Gretta and Price [26] yield the LDP protocols NBS and BayeSS [12], the latter with sample complexity $O(\log B/(\varepsilon^2\alpha^2))$ for rank error α over a domain of B values, optimal for small ε. Furthermore, Liu et al. [11] run Robbins–Monro stochastic approximation [27] with Polyak–Ruppert averaging [28] on threshold answers, and obtain confidence intervals through self-normalization [29], [30].

In this paper, we particularly target the adaptive protocols because each guarantees rank accuracy for every monotone response curve, which is the property our attack turns against it (§IV). Non-adaptive estimators instead read the quantile off a post-processed CDF estimate, and when the post-processing is non-linear, its effect on a poisoned CDF has no closed form; these estimators are also the targets of existing poisoning attacks [31]–[33]. We nevertheless cover the hierarchical histogram (HH), whose consistency step is linear, in §IV-E.

Poisoning attacks and defenses for LDP. Cheu et al. [34] showed that LDP protocols are inherently vulnerable to manipulation. Cao et al. [35] introduced targeted attacks on frequency estimation, followed by attacks on key–value data [36], mean and variance [37], numerical attributes [32], frequent itemsets [38], rankings [39], graphs [40], range queries [31], [33], trajectories [41], and streams [42]. Defenses include verifiable randomization [43], detection and recovery [44]–[46], differential aggregation against colluding attackers [47], and attack-agnostic robust estimation [48]. Most closely related, Li et al. [32] measure the robustness of numerical LDP protocols, whose shift toward the domain edge also inflates upper quantiles, but toward no chosen value; Liao et al. [31] show that post-processing amplifies poisoning of range queries, and Li et al. [33] design fine-grained manipulation attacks on range-query protocols. None of them targets threshold-query protocols. In statistics, Li et al. [13] derive minimax rates for median estimation under LDP when Huber contamination precedes privatization (our IPA setting), and Acharya et al. [49] bound manipulation attacks on distribution estimation under local constraints. Both bound the worst-case error under contamination. In contrast, we ask what budget places a tail quantile at a chosen value, an LDP analogue of the breakdown point [50], [51]: LDP with unverified reports asymptotically lowers it for upper quantiles.

III. Problem Formulation

We fix the data model, the class of protocols we attack, and the attacker's capabilities.

Data and quantiles. Each user holds a value in the ordered domain $[B] = \{1, \ldots, B\}$. Following [11], [12], honest values

are i.i.d. draws from a population distribution P on $[B]$ with CDF $F(t) = \Pr_{x\sim P}[x \le t]$ for $t \in \{0, 1, \ldots, B\}$, where $F(0) = 0$ and $F(B) = 1$. For a level $q \in (0, 1)$, the q-th quantile is $x_q = \min\{t \in [B] : F(t) \ge q\}$. A *tail quantile* has q close to 1 (e.g., $q = 0.99$).

Threshold-oracle protocols. A threshold query at $t \in \{0, \ldots, B\}$ asks a user holding x for $y = \mathrm{RR}_\varepsilon(\mathbf{1}[x \le t])$, the output of binary randomized response (§II) on the bit $\mathbf{1}[x \le t]$. Let $\rho_\varepsilon = 1/(1 + e^\varepsilon)$ denote the flip probability. The *response curve* $\pi(t) = \Pr[y = 1] = \rho_\varepsilon + (1 - 2\rho_\varepsilon)F(t)$ is non-decreasing in t, and the collector inverts it into the *debiased response curve*

$$G(t) = \frac{\pi(t) - \rho_\varepsilon}{1 - 2\rho_\varepsilon}, \qquad \text{so that } G = F \text{ for honest users.} \quad (1)$$

A protocol Π in the class $\mathcal{T}$ of *threshold-oracle protocols* contacts n users one at a time, sends each a threshold that may depend on all previous answers (adaptivity is allowed), and finally outputs an estimate $\hat{m} \in [B]$ of x_q. Each user is queried once, so Π is ε-LDP. In particular, a quantile protocol is designed to return the point where the response curve crosses level q. Following the noisy-binary-search literature [12], [25], [26], we state this guarantee for *arbitrary* monotone curves:

Definition 1 (Accuracy)**.** Let $\eta > 0$ and $\delta \in (0, 1)$. $\Pi \in \mathcal{T}$ is (η, δ)-*accurate at level* q if, for every non-decreasing curve $\pi' : \{0, \ldots, B\} \to [0, 1]$ with debiased curve G' defined as in (1), when every answer to a query at t is an independent Bernoulli$(\pi'(t))$ bit, the output satisfies (a) $G'(\hat{m}-1) < q+\eta$ unless $\hat{m} = 1$, and (b) $G'(\hat{m}) > q - \eta$ unless $\hat{m} = B$, with probability at least $1 - \delta$. Π is *robustly* (η, δ)-accurate if the same holds for every curve π', monotone or not.

Definition 1 is the "good coin" guarantee of noisy binary search over monotone coins [25], [26], translated from coin probabilities to debiased units. The exceptions for $\hat{m} \in \{1, B\}$ are the usual sentinel coins at the ends of the domain; they make the definition satisfiable for every curve, which needs $\eta > 0$. We study three protocols in $\mathcal{T}$: (i) NBS [12] splits the users into $\lceil \log_2 B \rceil$ batches; each batch estimates G at the midpoint of the current interval and halves it. (ii) BayeSS [12] runs the Bayesian screening search of [26] on randomized answers: it keeps a posterior over the intervals $[t, t+1]$, queries at the posterior median, updates multiplicatively, narrows the candidates to $O(1)$, and finishes with NBS. General quantiles are reduced to the median by padding with $(1 - q)n$ virtual users at 1 and qn at B. (iii) SGD-SN [11] asks each user whether x exceeds the current estimate, through a randomized comparison that is truthful with probability $r = \tanh(\varepsilon/2)$ and otherwise a fair coin (distributionally identical to RR_ε on $\mathbf{1}[x \le t]$). It then takes a Robbins–Monro step, Polyak–Ruppert averages the iterates, and reports a self-normalized 95% confidence interval (CI); we evaluate a log-scale variant, SGD-SN-log (§VI-A). NBS is robustly $(\eta_{\mathrm{NBS}}, \delta)$-accurate with $\eta_{\mathrm{NBS}} = \sqrt{\ln(2R/\delta)/(2k)}\,/(1 - 2\rho_\varepsilon)$, where $R = \lceil \log_2 B \rceil$ and $k = \lfloor n/R \rfloor$: by Hoeffding's inequality and a union bound, every batch estimate is within η_{NBS} of G' with probability at least $1 - \delta$, for any curve and adaptively chosen thresholds, and the search keeps an estimate below q just left of its interval and one at or above q at its right end without using monotonicity. Holding for every curve and round at once, this bound is conservative; the calibrated tolerances of Remark 1 (see later) capture the much smaller error of a typical run. The analysis of BayeSS likewise applies to any non-decreasing sequence of coin biases [12], [26] and locates a crossing interval with error $\tilde{O}\big(\sqrt{\log B/n}/(1-2\rho_\varepsilon)\big)$; we return the first queried threshold whose padded estimate reaches $1/2$, which approximates the output convention of Definition 1 (queried thresholds need not be contiguous). The padding maps level q to $1/2$ and halves every margin. We derive no explicit constants for BayeSS's schedule, so its η, unlike η_{NBS}, is not certified, and its results are empirical. SGD-SN-log is analyzed for continuous data with a positive density at the quantile [11], which neither $[B]$ nor a step-shaped curve provides; our theorems do not cover it, and we evaluate it empirically. As a non-adaptive reference outside $\mathcal{T}$, we also evaluate the hierarchical histogram (HH) [9].

Threat model. The attacker controls fake users, e.g., emulated devices. As in Huber's contamination model [50], every user contacted by the protocol is fake independently with probability $\beta \in [0, 1)$, the *poisoning rate*; the collector enforces this by contacting users drawn without replacement from a registered pool much larger than n (the independent model is its limit, and Hoeffding's bound holds without replacement) and assigning them to rounds at random, so fake accounts cannot choose when they answer.

More specifically, we consider the *output poisoning attack* (OPA) [35], [37]: a fake user may bypass the randomizer and send any answer. A strategy is a function $s : \{0, \ldots, B\} \to [0, 1]$, where $s(t)$ is the probability that a fake user answers 1 to a query at t, with randomness independent of all other users. On the other hand, under the *input poisoning attack* (IPA), fake users run the honest randomizer on a value of their choice, which may depend on the query. The attacker knows the protocol, n, ε, q, and B, all of which are public, but not the honest data. Its goal is *targeted inflation*: for a chosen $m^\star > x_q$, make the protocol report $\hat{m} = m^\star$ (e.g., to raise an alerting threshold above the attacker's own activity by a chosen amount rather than to the domain edge). We call $\kappa = m^\star/x_q$ the *inflation factor* and measure attack cost by β.

Poisoned response curve. Under a strategy s, a query at t is answered 1 with probability $\pi_\beta(t) = (1 - \beta)\pi(t) + \beta s(t)$, so the debiased curve becomes

$$G_\beta(t) = (1 - \beta)F(t) + \beta g(t), \qquad g(t) = \frac{s(t) - \rho_\varepsilon}{1 - 2\rho_\varepsilon}. \quad (2)$$

The constant

$$c_\varepsilon = \frac{\rho_\varepsilon}{1 - 2\rho_\varepsilon} = \frac{1}{e^\varepsilon - 1} \quad (3)$$

will measure how far debiasing stretches a single report: $c_\varepsilon = 0.58$ at $\varepsilon = 1$ and 1.54 at $\varepsilon = 0.5$.

IV. Tail Saturation and the Cliff Attack

Here, we bound a fake user's influence, show why the tail is cheap to move, and build the attack.

A. Influence of a fake user

A fake user acts only through its debiased answers, so we first bound what one answer can contribute.

Lemma 1 (Influence envelope)**.** *For every OPA strategy, poisoning rate β, and threshold t,*

$$(1-\beta)F(t) - \beta c_\varepsilon \;\le\; G_\beta(t) \;\le\; (1-\beta)F(t) + \beta(1+c_\varepsilon).$$

Under IPA, it narrows to $[(1-\beta)F(t), (1-\beta)F(t)+\beta]$.

Proof sketch: In Eq. (2), $s(t) \in [0,1]$ gives $g(t) \in [-c_\varepsilon, 1+c_\varepsilon]$. Under IPA, $s(t) \in [\rho_\varepsilon, 1-\rho_\varepsilon]$ gives $g(t) \in [0,1]$. Full proofs of all results are in the appendix. ∎

A fake "no" thus counts as $-c_\varepsilon$, whereas an honest user contributes $\mathbf{1}[x \le t] \in \{0,1\}$ in expectation. This debiasing amplification, also exploited for frequency estimation [35], grows as ε shrinks. Another randomizer cannot avoid it:

Lemma 2 (RR_ε amplifies least)**.** *Let a binary ε-LDP randomizer answer 1 with probability a if $x \le t$ and $b < a$ otherwise, and debias an answer y to $(y-b)/(a-b)$. A reported 0 then counts at most $-c_\varepsilon$ and a reported 1 at least $1+c_\varepsilon$, and both bounds are attained simultaneously if and only if the randomizer is RR_ε.*

Proof: ε-LDP gives $a \le e^\varepsilon b$ and $1-b \le e^\varepsilon(1-a)$, i.e., $b/(a-b) \ge c_\varepsilon$ and $(1-a)/(a-b) \ge c_\varepsilon$. ∎

Other randomizers thus only widen the envelope, so Theorem 1 and Corollary 1 still hold for them.

B. Tail saturation and the rank–value gap

A threshold-oracle protocol locates x_q through the sign of $G(t) - q$ at the thresholds it queries. To make it report $m^\star$, the attacker must reverse that sign at every queried threshold in $[x_q, m^\star)$, and nowhere else. Each reversal costs the honest margin there, which the tail keeps small:

Observation 1 (Tail saturation)**.** For every $t \ge x_q$, $F(t) \in [q, 1]$. Hence, the honest margin $G(t) - q$ of every threshold decision above the quantile is at most $1-q$, independently of how far t lies from x_q.

By Observation 1, each reversal costs at most $1-q$ in debiased units, whether $m^\star = 1.1\,x_q$ or $m^\star = B$. The cost is therefore set by the rank $F(m^\star - 1)$; the value $m^\star$ is free. This is the rank–value gap. The gap holds for every quantile, but it is dangerous in the tail, where $1-q$ is small: for the median, a reversal far above $x_{0.5}$ costs up to $1/2$. For the p99 at $\varepsilon = 1$, a reversal thus costs at most one point of debiased mass at any distance from x_q, and 0.63% of users suffice asymptotically.

Algorithm 1 Cliff

Require: public protocol Π with n, ε, q, $[B]$; target $m^\star$ and guessed inflation $\hat\kappa$ ($\hat\kappa \gg 1$ without knowledge of x_q); success level $1-\delta$; margin μ; (optional) flag Stealthy with $L \le x_q$

Offline: calibrate the rate and step (no data needed)

1: $\tilde P \leftarrow$ synthetic population on $[B]$ with q-quantile $m^\star/\hat\kappa$; $m_s \leftarrow m^\star$
2: **for** candidate rates β in increasing order **do**
3: **if** Π reports an average of its iterates **then** $m_s \leftarrow \lambda m^\star$, where $\lambda = m^\star/\tilde m$ for the output $\tilde m$ of Π simulated on $\tilde P$ with a step at $m^\star$
4: **if** Π, simulated on $\tilde P$ with a step at m_s, outputs $m^\star$ w.p. $\ge 1-\delta$ **then break** ▷ or Remark 1 with worst-case ranks
5: **end for**
6: $\beta \leftarrow (1+\mu)\beta$ ▷ costs budget, not precision: it only deepens the cliff

Online: a fake user's answer to a query at threshold t

7: **if** Stealthy **and** $t \notin [L, m_s)$ **then**
8: **return** $\mathrm{RR}_\varepsilon(\mathbf{1}[m_s \le t])$ ▷ behave like an honest user
9: **end if**
10: **return** $\mathbf{1}[t \ge m_s]$

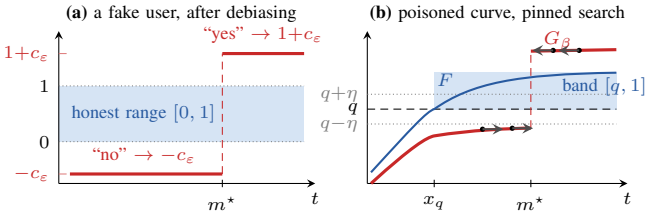


Fig. 2. Cliff in one picture. (a) A fake user answers "no" below $m^\star$ and "yes" from $m^\star$ on, without noise; debiasing maps them to $-c_\varepsilon$ and $1+c_\varepsilon$, the extremes of Lemma 1. (b) G_β stays below $q-\eta$ on $[x_q, m^\star)$ and jumps above $q+\eta$ at $m^\star$, so an accurate protocol moves right at every probe below $m^\star$ and left from $m^\star$ on: it outputs $m^\star$.

C. The Cliff attack

Cliff turns tail saturation into an attack: every fake user answers as a noise-free holder of a step position m_s, so the debiased curve drops below q just before m_s and jumps above it at m_s (Fig. 2). Algorithm 1 has two parts: an offline calibration of the rate and the step without honest data (lines 1–6), and the online answer of a fake user to a threshold query (lines 7–10). In particular, without Stealthy, lines 7–10 of Algorithm 1 set $s(t) = \mathbf{1}[t \ge m_s]$: fake users behave like users holding the step position m_s, but never flip their answers. With Stealthy (S-Cliff), fake users stay noise-free only on $[L, m_s)$, where decisions must be reversed, and elsewhere randomize like honest users holding m_s: every debiased estimate then stays in $[0,1]$ whenever $F(L) > \beta c_\varepsilon/(1-\beta)$, which hides the attack from range and boundary audits, at the price of a smaller jump at $m^\star$, $\beta(1+c_\varepsilon)$ instead of $\beta(1+2c_\varepsilon)$. We analyze the step at $m_s = m^\star$. By

Eq. (2),

$$G_\beta(t) = \begin{cases} (1-\beta)F(t) - \beta c_\varepsilon, & t < m^\star, \\ (1-\beta)F(t) + \beta(1+c_\varepsilon), & t \geq m^\star. \end{cases} \quad (4)$$

Below $m^\star$, the curve sits on the lower envelope of Lemma 1. From $m^\star$ on, it sits on the upper envelope, so it jumps by at least $\beta(1+2c_\varepsilon)$ at $m^\star$ (Fig. 2). An accurate protocol must stop where the curve crosses level q, so a deep enough cliff fixes its output:

Theorem 1 (Pinning)**.** *Let $\Pi \in \mathcal{T}$ be (η,δ)-accurate at level q, and let $m^\star \in [B]$ satisfy $F(m^\star) \geq q+\eta$ or $m^\star = B$. If*

$$\beta \;\geq\; \beta^\star_\eta(m^\star) := \frac{F(m^\star-1) - q + \eta}{F(m^\star-1) + c_\varepsilon}, \quad (5)$$

then under CLIFF$(m^\star)$*, Π outputs $\hat{m} = m^\star$ with probability at least $1-\delta$.*

Proof sketch: By (4) and monotonicity, $G_\beta(t) \leq (1-\beta)F(m^\star-1) - \beta c_\varepsilon \leq q - \eta$ for all $t < m^\star$, where the last step is (5). If $m^\star < B$, then $G_\beta(t) \geq F(m^\star) \geq q+\eta$ for $t \geq m^\star$. The poisoned curve is non-decreasing, so Definition 1 applies to it. Any $\hat{m} < m^\star$ (hence $\hat{m} < B$) violates condition (b), and any $\hat{m} > m^\star$ (hence $\hat{m} > 1$; none if $m^\star = B$) violates (a). ∎

Theorem 1 turns the protocol's correctness against it: with probability $\geq 1-\delta$, it stops exactly at the cliff, not "somewhere in the tail." Raising β above $\beta^\star_\eta$ only deepens the cliff, so the attacker can over-provision without overshooting, unlike attacks without a step at $m^\star$ such as MGA. The appendix walks through a toy example. Since Eq. (5) grows with the unknown rank $F(m^\star-1)$, its worst case covers all targets:

Corollary 1 (Universal budget)**.** *For $\eta < q + c_\varepsilon$ and every target $m^\star$ with $F(m^\star) \geq q+\eta$ or $m^\star = B$,*

$$\beta^\star_\eta(m^\star) \;\leq\; \beta^\infty_\eta := \frac{1-q+\eta}{1+c_\varepsilon} = (1-q+\eta)\,(1-e^{-\varepsilon}).$$

Proof sketch: $(F-q+\eta)/(F+c_\varepsilon)$ is increasing in $F \leq 1$, and $1/(1+c_\varepsilon) = 1-e^{-\varepsilon}$. ∎

We write $\beta^\star = \beta^\star_0(m^\star) = \lim_{\eta\to 0}\beta^\star_\eta(m^\star)$ for the asymptotic threshold. Any rate above $(1-q)(1-e^{-\varepsilon})$ lets the attacker place the q-th quantile anywhere in $(x_q, B]$ once η is small enough, i.e., for large n (by Remark 1 also where $F(m^\star) < q+\eta$), without knowing the data beyond $m^\star > x_q$. For the p99, this is 0.63% of users at $\varepsilon = 1$ and 0.39% at $\varepsilon = 0.5$. It is sufficient for every target but not necessary: a target with $F(m^\star-1) < 1$ costs less. At finite n, the condition $F(m^\star) \geq q+\eta$ of Theorem 1 can fail, and the attacker needs a data-free budget:

Remark 1 (Finite-sample budget)**.** The proof of Theorem 1 needs only $G_\beta(t) \leq q-\eta$ for $t < m^\star$ and $G_\beta(t) \geq q+\eta$ for $t \geq m^\star$. Under CLIFF, the second condition holds once $\beta \geq (q+\eta-F(m^\star))/(1+c_\varepsilon-F(m^\star))$, even if $F(m^\star) < q+\eta$, so the certified budget at finite n is the larger of this value and $\beta^\star_\eta(m^\star)$. The same argument certifies a window: with $\ell = \lceil 0.9m^\star \rceil > 1$ and $u = \min\{\lfloor 1.1m^\star \rfloor, B\}$, the output lies in $[\ell, u]$ once $\beta \geq (F(\ell-1)-q+\eta)/(F(\ell-1)+c_\varepsilon)$ and, if $u < B$, $\beta \geq (q+\eta-F(u))/(1+c_\varepsilon-F(u))$, the hit criterion of §VI. Under IPA, $G_\beta \leq 1$, so once $\eta > 1-q$, condition (a) holds everywhere and the argument can no longer pin a target below B. Because Definition 1 quantifies over all monotone curves, η depends on the protocol, n, ε, B, q, and δ, but not on the data; for NBS, η_{NBS} is known without data, so Theorem 1 with this relaxation certifies a budget from the target's rank, and $\max\{(1-q+\eta)(1-e^{-\varepsilon}), \eta/(1+c_\varepsilon-q)\}$ for every target above x_q. Tighter, *empirical* tolerances $\hat{\eta}_-$ (below the target) and $\hat{\eta}_+$ (above it) come from running the protocol on synthetic step curves (§VI-A); unlike the distribution-free η of Definition 1, they predict the budget but guarantee nothing. The certified budget depends on the target only through its ranks, the actual one also on the search path (appendix).

Calibration without data. The attacker sets the rate β and the step position m_s offline (Algorithm 1, lines 1–6). Because the cost depends mainly on rank, it simulates the public protocol on a synthetic population on the public domain, or evaluates Remark 1 with $\hat{\eta}_\pm$ and worst-case ranks, $\max\{(1-q+\hat{\eta}_-)(1-e^{-\varepsilon}), \hat{\eta}_+/(1+c_\varepsilon-q)\}$. An overestimate of κ is conservative below the target, but the synthetic population fixes the rank only below $m^\star$; data whose CDF stays near q above $m^\star$ need the two-sided bound. Line 3 is designed for SGD-SN-log: a protocol that reports an average of its iterates lags behind the step, so the attacker places the step ahead of the target, at $\lambda m^\star$ with $\lambda > 1$ (details in RQ2). All else comes from the public protocol.

D. Optimality and the price of local privacy

Theorem 1 shows that $\beta^\star_\eta$ suffices. Could a strategy with noisy or graded answers need far fewer fake users? Among time-invariant strategies, no: Lemma 1 caps every strategy at every threshold, so $\beta^\star_\eta$ is also necessary up to $O(\eta)$.

Theorem 2 (Optimality of CLIFF)**.** *Let $\Pi \in \mathcal{T}$ be (η,δ)-accurate at level q, and consider any time-invariant OPA strategy s that keeps the poisoned curve non-decreasing, or any time-invariant strategy if Π is robustly accurate. If $\beta < (F(m^\star-1)-q-\eta)/(F(m^\star-1)+c_\varepsilon)$, then Π outputs $\hat{m} < m^\star$ with probability at least $1-\delta$.*

Proof sketch: By Lemma 1, $G_\beta(t) \geq (1-\beta)F(m^\star-1) - \beta c_\varepsilon > q+\eta$ for all $t \geq m^\star - 1$, so condition (a) forces $\hat{m} - 1 < m^\star - 1$ or $\hat{m} = 1 < m^\star$. ∎

For targets with $F(m^\star) \geq q+\eta$, CLIFF is thus optimal up to an additive $2\eta/(F(m^\star-1)+c_\varepsilon)$, i.e., asymptotically. The gap shrinks as n grows. For NBS at $n = 10^6$, however, $\eta_{\mathrm{NBS}} > 1-q$ (this holds up to $n \approx 2.8\times 10^6$), so the lower bound is vacuous, and tail targets are certified only through Remark 1. Strategies that change over time face the same envelope at every query. For NBS, whose rounds use fresh users, the per-round Hoeffding bound of η_{NBS} lets the argument go through unchanged. A coupling shows even more for NBS: at every n and β, no OPA strategy, time-varying or coordinated, makes it output exactly $m^\star$ more often than CLIFF (appendix). To isolate what local privacy adds, we compare with IPA, whose fake users must run the honest randomizer:

Proposition 1 (Price of local privacy)**.** *Under IPA,* CLIFF*'s honest-randomizer analogue (fake users holding $m^\star$) pins $m^\star$, where $F(m^\star) \geq q+\eta$, whenever $\beta \geq \beta_\eta^{\mathrm{IPA}}(m^\star) := (F(m^\star-1)-q+\eta)/F(m^\star-1)$, and no IPA strategy that keeps the curve non-decreasing makes Π output $\hat{m} \geq m^\star$ if $\beta < (F(m^\star-1)-q-\eta)/F(m^\star-1)$. As $\eta \to 0$, $\beta_\eta^{\mathrm{IPA}}(m^\star)$ tends to the contamination rate beyond which the non-private quantile moves to $m^\star$. For every target with $F(m^\star-1) > q$, the asymptotic thresholds satisfy*

$$\frac{\beta_0^\star(m^\star)}{\beta_0^{\mathrm{IPA}}(m^\star)} = \frac{F(m^\star-1)}{F(m^\star-1)+c_\varepsilon} \;\leq\; 1-e^{-\varepsilon},$$

with equality if and only if $F(m^\star-1)=1$; at finite η the certified IPA budget can be infinite (Remark 1).

Proof sketch: Under IPA, Lemma 1 confines g to $[0,1]$; the proofs of Theorems 1 and 2 hold with $c_\varepsilon = 0$. ∎

Without privacy, the upward breakdown of the q-th quantile is $1-q$ [51]. LDP with unverified reports (OPA) asymptotically lowers it to $(1-q)(1-e^{-\varepsilon})$, and, while η is small, the stronger the privacy, the cheaper the attack, as for frequency estimation [34], [35]. Verifiable randomization [43], which turns OPA into IPA, raises the asymptotic cost by $1+c_\varepsilon/F(m^\star-1)$, at least $1/(1-e^{-\varepsilon})$ and close to it for tail targets; even then the asymptotic budget stays at most $1-q$. An attacker may instead *lower* a tail quantile, e.g., to hide a latency regression; this costs more:

Proposition 2 (Direction asymmetry)**.** *To deflate the reported quantile to $m^\star < x_q$ with $F(m^\star) \leq q-\eta$, the strategy $s(t) = \mathbf{1}[t \geq m^\star]$ succeeds whenever $\beta \geq (q+\eta-F(m^\star))/(1+c_\varepsilon-F(m^\star))$, and no time-invariant strategy that keeps the curve non-decreasing succeeds if $\beta < (q-\eta-F(m^\star))/(1+c_\varepsilon-F(m^\star))$. Consider a rank distance $d>0$, an inflation target with $F(m^\star-1)=q+d$, and a deflation target with $F(m^\star)=q-d$. As $\eta \to 0$, deflation costs $(q+d+c_\varepsilon)/(1-q+d+c_\varepsilon)$ times more than inflation, a ratio that tends to e^ε as $q \to 1$ and $d \to 0$.*

Proof sketch: Mirror Theorems 1 and 2: $s = \mathbf{1}[t \geq m^\star]$ lifts G to $(1-\beta)F+\beta(1+c_\varepsilon)$ from $m^\star$ on, and the envelope bounds every strategy; substituting $F = q \pm d$ gives the ratio. ∎

Raising a tail threshold is therefore much cheaper than lowering it, and it is also what an attacker who wants to evade a threshold needs.

E. Beyond threshold oracles

The hierarchical histogram HH [9] assigns each user to one level of a tree over $[B]$; the user sends an optimized unary encoding (OUE) [15] of the node that contains its value, i.e., a bit per node that is 1 with probability $p_o = \frac{1}{2}$ for its own node and $q_o = 1/(e^\varepsilon+1)$ for every other node. The collector debiases each node count, post-processes the tree for consistency [18] (fixing the root at 1), and reports the first value at which the prefix sum of the leaves reaches q (binary search over the tree [9] can differ where noisy prefixes are non-monotone). Generic CLIFF, a noise-free one-hot vector for the node of $m^\star$, lowers every other node of every level by the same amount. These shifts are not consistent across levels, so the post-processing absorbs most of them for targets below B. The remedy is to inject a consistent shift:

Proposition 3 (Tree-consistent CLIFF)**.** *Let HH use any affine post-processing that fixes the root at 1 and leaves consistent trees unchanged. Let each fake user report, at its level, bit 1 with probability 0 for the node that contains value 1, p_o+q_o for the node that contains $m^\star$, p_o if one node contains both, and q_o for every other node. Then, the expected post-processed prefix sum is $(1-\beta)F(t)-2\beta c_\varepsilon$ for $1 \leq t < m^\star$ and $(1-\beta)F(t)+\beta$ for $t \geq m^\star$. If, with probability at least $1-\delta$, every prefix sum deviates from its expectation by less than η, then HH outputs $m^\star$ whenever $F(m^\star) \geq q+\eta$ or $m^\star = B$, and $\beta \geq (F(m^\star-1)-q+\eta)/(F(m^\star-1)+2c_\varepsilon)$, for $\eta < q+2c_\varepsilon$ a budget of at most $(1-q+\eta)\tanh(\varepsilon/2)$.*

Proof sketch: A bit with probability s debiases to $(s-q_o)/(p_o-q_o)$, which equals $-2c_\varepsilon$, $1+2c_\varepsilon$, 1, and 0 for the four cases. The fake users therefore add, at every level, the node sums of the signed measure $-2c_\varepsilon\,\mathbf{u}_1 + (1+2c_\varepsilon)\,\mathbf{u}_{m^\star}$, where $\mathbf{u}_v$ is the unit point mass at v; it has total mass 1. The expected tree is consistent with root 1, so the post-processing leaves it unchanged, and its prefix sums are as stated. The rest follows the proof of Theorem 1. ∎

We call this attack T-CLIFF. A fake user removes mass at a small value and adds it at $m^\star$, a consistent change that post-processing cannot undo. Spreading the removed mass uniformly over $[1,L]$ leaves every prefix sum on $[L,m^\star)$ unchanged, so Proposition 3 still applies if also $(1-\beta)F(L-1) \leq q-\eta$, and lowers node v by $2\beta c_\varepsilon|v \cap [1,L]|/L$: the full $2\beta c_\varepsilon$ falls only on nodes that cover $[1,L]$ and thus also carry its honest mass $F(L)$. Because OUE debiases a fake 0 to $-2c_\varepsilon$ rather than $-c_\varepsilon$, T-CLIFF's asymptotic budget, an upper bound for HH, is lower than that of threshold-oracle protocols (0.46% against 0.63% for the p99 at $\varepsilon=1$). Like IPA, however, T-CLIFF lifts the curve above $m^\star$ by only $\beta(1-F)$, so the condition $F(m^\star) \geq q+\eta$ requires $\eta \leq 1-q$; Proposition 3 therefore becomes vacuous for $m^\star < B$ as HH's error approaches $1-q$ (T-CLIFF is not claimed optimal), as it nearly does at $n = 10^7$. Generic CLIFF is the baseline tree attack of [31] for the point range $\{m^\star\}$ without its random padding; for the range $[m^\star, B]$, this baseline, which we call RQA (range-query attack), sets every node from $m^\star$ on, is not cancelled, and overshoots to the domain edge.

V. DEFENSE: CERTIFIED TAIL QUANTILES

Even without privacy, the upward breakdown point of the q-th quantile is $1-q$, against $1/2$ for the median, so no estimator that is consistent on clean data makes the tail as robust as the median. Proposition 1 shows that LDP lowers it further and, by Lemma 2, no binary randomizer avoids this. On the other hand, the defender can change what it estimates. Our defense, CTQ, assumes an upper bound $\bar{\beta}$ on the probability that a responding user is fake (for NBS, a pool of N accounts with N_f fake, all responding, gives $\bar{\beta} = N_f/(N-n)$, and the

honest CDF drifts by at most $n/(N - N_f - n)$, which adds to η), and runs an accurate protocol at a level lowered enough that no admissible attack can push the output above x_q:

Theorem 3 (CTQ certificate)**.** *Let $\Pi \in \mathcal{T}$ be (η, δ)-accurate at level $q^\dagger = q - \bar{\beta}(q + c_\varepsilon) - \eta > 0$. For every time-invariant OPA strategy with $\beta \le \bar{\beta}$ that keeps the poisoned curve non-decreasing, and for every one if Π is robustly accurate (e.g., NBS), with probability at least $1 - \delta$,*

$$x_{q^\ddagger} \;\le\; \hat{m} \;\le\; x_q, \qquad q^\ddagger = \frac{q - \bar{\beta}(1 + q + 2c_\varepsilon) - 2\eta}{1 - \bar{\beta}},$$

where $x_{q'} := 1$ for $q' \le 0$.

Proof sketch: For $t \ge x_q$, Lemma 1 gives $G_\beta(t) \ge (1-\beta)q - \beta c_\varepsilon \ge (1-\bar{\beta})q - \bar{\beta}c_\varepsilon = q^\dagger + \eta$, so condition (a) at level $q^\dagger$ forces $\hat{m} \le x_q$. For t with $F(t) < q^\ddagger$, the upper envelope gives $G_\beta(t) < (1-\bar{\beta})q^\ddagger + \bar{\beta}(1 + c_\varepsilon) = q^\dagger - \eta$, so condition (b) forces $\hat{m} > t$. Both steps use the envelope pointwise and never the monotonicity of G_β. ∎

CTQ pays with a downward bias on clean data, and no certified estimator of this kind avoids it:

Proposition 4 (Necessity of the shift)**.** *Let $\mathcal{A}$ be any estimator that obtains from each user one answer to a threshold query through RR_ε or any other binary ε-LDP randomizer (possibly depending on the threshold), with thresholds chosen adaptively or not and arbitrary post-processing (accurate or not). Suppose that $\mathcal{A}$ outputs $\hat{m} \le x_q$ with probability at least $1 - \delta$ for every population and every time-invariant OPA strategy with $\beta \le \bar{\beta}$ that keeps the (debiased) poisoned curve non-decreasing. Then on every clean population, $\mathcal{A}$ outputs a value at most $x_{q-\bar{\beta}(q+c_\varepsilon)}$ with probability at least $1-\delta$. Against IPA, the necessary shift is $\bar{\beta}q$.*

Proof sketch: Given a clean population with CDF F, define $F'(t) = \min\{1, (F(t)+\bar{\beta}c_\varepsilon)/(1-\bar{\beta})\}$. An OPA strategy with $g(t) = (F(t) - (1-\bar{\beta})F'(t))/\bar{\beta} \in [-c_\varepsilon, 1]$, achievable under any binary randomizer by Lemma 2, makes the poisoned curve of F' coincide with the clean curve of F, so every transcript has the same distribution in both worlds, and the true q-quantile of F' is the $(q - \bar{\beta}(q + c_\varepsilon))$-quantile of F. ∎

Among estimators that collect one randomized threshold answer per user, CTQ is therefore optimal up to $O(\eta)$: no distribution-free certifiably inflation-proof estimator avoids a downward shift of $\bar{\beta}(q + c_\varepsilon)$ in rank. For NBS, the per-round argument after Theorem 2 extends the certificate to strategies that change over time. If fake accounts could choose their round, $\bar{\beta}$ would have to bound the fake fraction of every round, not the account rate. At $\varepsilon = 1$ and $\bar{\beta} = 1\%$, this shift alone is 1.6 points of rank; η adds to it ($\eta_{\mathrm{NBS}} = 1.7$ points at $n = 10^6$).

VI. Evaluation

We answer six research questions: **RQ1** (mechanism) Does the attack cost depend on rank rather than on how far the quantile moves? **RQ2** (protocol) Which protocol designs are fragile, and on which data? **RQ3** (attacker) How much knowledge and budget precision does the attacker need, and why is the cliff necessary? **RQ4** (boundary) When does the attack become expensive? **RQ5** (defender) Can audits catch CLIFF, and what does CTQ cost? **RQ6** (consequence) What happens downstream?

TABLE I
DATASETS: TRUE P99 (RAW UNIT AND VALUE IN $[B]$), HEADROOM, AND EFFECTIVE INFLATION κ_{eff} OF THE DEFAULT TARGET.

dataset	records	B	raw	$x_{0.99}$	$B/x_{0.99}$	κ_{eff}	tail
CreditCard	284,807	2^{20}	1,018 EUR	20,361	51.5	10	heavy
Flights	200,000	2^{12}	137 min	224	18.3	10	moderate
Power	2,049,280	2^{14}	4,850 W	4,851	3.4	3.38	bounded
Pareto(1.5)	2,000,000	2^{20}	40,778	40,779	25.7	10	heavy

A. Experimental setup

Datasets. Table I lists the three real datasets and the synthetic Pareto population used throughout RQ2. *CreditCard*[1] contains 284,807 card transactions, 0.17% of them labeled fraudulent; the attribute is the amount, in units of 0.05 EUR. *Flights*[2] contains 200,000 U.S. flight arrival delays in minutes. We add 87 minutes so that the earliest arrival maps to value 1 (ranks are unchanged; $10\times$ the shifted p99 is $15.7\times$ the p99 delay). *Power*[3] contains 2,049,280 minute-level readings of household active power in watts. The three datasets span heavy (CreditCard), moderate (Flights), and bounded (Power) tails. Seven synthetic distributions on $B = 2^{20}$ vary the tail: Pareto (Lomax, i.e., Pareto II) with shape $a \in \{1.1, 1.5, 2.5\}$ scaled by 2000 as in [12], lognormal with $\mu = 8$ and $\sigma \in \{1, 2\}$, exponential with mean 5000, and uniform on $[0, 0.8B]$. As in [9], [11], [12], each dataset defines a population, and users are drawn i.i.d. from it. Values are clipped to $[B]$. The public domain B is a power of four (as HH requires) covering each attribute's plausible range.

Protocols and parameters. NBS and BayeSS follow the reference implementation of [12], except for BayeSS's output rule (§III). BayeSS uses the update constant $\min\{0.24, 0.6\sqrt{\ln B/M}\}$ (M: padded population size) and the schedule of [12], with an equivalent $O(\log B)$ lazy segment tree for its weight update [26]. SGD-SN-log uses the step size $d_k = 2/(k^{0.51} + 100)$ of [11] for its k-th user, with the self-normalized critical value 6.747 [30]. We run its iterates θ_k on the $\log_2$ scale, starting at $\frac{1}{2}\log_2 B$, and ask user k whether $x \le 2^{\theta_k}$; answers and quantiles are unchanged, the step dynamics differ, and the Polyak–Ruppert average becomes a geometric mean, which avoids tuning a raw-scale step size per domain; SGD-SN-log reports $\lceil 2^{\bar{\theta}_n} \rceil$ for its averaged iterate $\bar{\theta}_n$. HH uses fan-out 4, optimized unary encoding [15], uniform level sampling, and the consistency step of [18], and reports the first prefix whose estimate reaches q, which biases its clean output low (Table III). Unless stated otherwise,

[1] https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

[2] https://github.com/vega/vega-datasets (`flights-200k`), derived from U.S. Bureau of Transportation Statistics records.

[3] UCI Individual Household Electric Power Consumption, https://archive.ics.uci.edu/dataset/235.

$\varepsilon = 1$, $q = 0.99$, $\kappa = 10$ (the target $m^\star = \min\{10x_q, B\}$; $\kappa_{\text{eff}} = m^\star/x_q = 3.38$ on Power), and $n = 10^6$; HH uses $n = 10^7$ because it is far less accurate at $n = 10^6$, as [12] also found. Across experiments, we vary $\varepsilon \in \{0.25, 0.5, 1, 2, 4\}$, $q \in \{0.5, 0.75, 0.9, 0.95, 0.99, 0.999\}$, $\kappa \in \{2, 5, 10, 50\}$, $n \in [10^5, 10^8]$ (10^9 in one audit), and β from 0 to 5% in fixed-budget runs (up to 40% in Table II).

Attacks. Besides CLIFF, its stealthy variant S-CLIFF (Algorithm 1 with STEALTHY, $L = x_q/2$), CLIFF with the shifted step of line 3 against SGD-SN-log, and T-CLIFF against HH, we adapt the main existing poisoning attacks to the quantile target. MGA, the maximal-gain attack of [35], sends noise-free reports crafted for the attacker's target items; CLIFF instantiates it at $m^\star$, and the untargeted instance that maximizes the gain answers as a holder of the largest value B, our MGA baseline. IPA, the input poisoning of [34], [37], runs the honest randomizer on $m^\star$. The fine-grained output attack of [37] knows the honest CDF F and solves for the fake answers that make the expected estimate equal to a target CDF H whose q-quantile is $m^\star$, i.e., each fake user contributes $g_H = (H - (1-\beta)F)/\beta$, clipped to the feasible $[-c_\varepsilon, 1 + c_\varepsilon]$. FGA-R relocates the honest mass of $[x_q, m^\star)$ to $m^\star$; FGA-C takes H as CLIFF's expected curve clipped to $[0, 1]$, for which g_H is always feasible. Against HH, RQA, the baseline tree attack of [31], inflates the estimated mass of the range $[m^\star, B]$ by setting every node from $m^\star$ on. RANDOM, whose fake users answer every query with a fair coin (a debiased contribution of $1/2$ at every threshold), is a naive reference: at $3\beta^\star$, it inflates the p99 $24\times$ on average under NBS but hits the target in at most 5% of runs (appendix).

Metrics. A trial *hits* the target if $|\hat{m} - m^\star| \le 0.1\, m^\star$ and *reaches* it if $\hat{m} \ge 0.9\, m^\star$. We report hit rates, achieved inflation $\hat{m}/x_q$, the rank $F(\hat{m})$, and exact landing ($\hat{m} = m^\star$); reach rates, used for one-sided goals in RQ3, are tabulated in the extended version (appendix). The *empirical budget* $\hat{\beta}$ estimates the smallest β with hit rate ≥ 0.9 by a noisy bisection (13 steps of 40 trials in Table II). We run 40 trials per configuration (20 for BayeSS; other counts are stated where used), with fixed seeds. Our code is available at https://github.com/chiamuyu/LDP-quantile-poisoning.

B. RQ1: Does the cost follow the theory?

Displacement independence. Fig. 3(a,b) varies the target from $2x_q$ to $50x_q$. The theoretical budgets are 0.49%–0.63% for $\kappa = 2$–50 (0.44%–0.63% for the $\pm 10\%$ window): a 25-fold larger target costs only 1.3–$1.4\times$ more, and none exceeds the universal bound 0.63% (Corollary 1). Empirically, the 90%-hit points of NBS and BayeSS lie right of the dashed lines by about the protocol error (at the last grid point left of them, hit rates are 0.05–0.48: at finite n, protocol noise also helps the attacker), their 90%-hit budgets grow only about $1.5\times$ from $\kappa = 2$ to 50 (NBS: 0.77% to 1.12%; BayeSS: 0.69% to 1.00%), and all reach a hit rate of 0.95 by $\beta = 1.2\%$; at $\beta = 1\%$, BayeSS hits $10x_q$ in 95% of runs and NBS in 88%. SGD-SN-log (Fig. 3(c)) is slower: its Polyak–Ruppert average includes iterates from before the attack took hold, so at $\beta = 1\%$ it reaches only $7.7 \times x_q$ for $\kappa = 10$ and needs about 2% (bisection: 1.87%, Fig. 3(d)). Its lag grows with the distance to the target ($\kappa = 2$: hit rate 0.95 at 0.8%; $\kappa = 50$: 0.48 at 3%), so its cost also depends on value.

Cost law and price of privacy. Table II compares $\hat{\beta}$ with two references. "thm." is the window bound of Remark 1 as $\eta \to 0$, $\beta_0^\star(\lceil 0.9m^\star \rceil)$ (β_0^{IPA} for IPA), within 0.02 points of $\beta_0^\star(m^\star)$ for $\kappa = 10$ and $q \ge 0.95$. "pred." evaluates Remark 1 at the hit window and the true ranks of the target with tolerances calibrated without data: every user answers from a step curve that is 0 below $m^\star/10$, $q - \Delta$ on $[m^\star/10, m^\star)$, and $1 + c_\varepsilon$ from $m^\star$ on, and $\hat{\eta}_-$ is the smallest Δ for which the protocol hits $m^\star$ in 90% of runs; $\hat{\eta}_+$ uses $-c_\varepsilon$ on $[m^\star/10, m^\star)$ and $q + \Delta$ from $m^\star$ on. At $n = 10^7$, the finite-n prediction matches CLIFF's budget to within 0.04 points in every cell; since CLIFF's curve near a $\kappa = 10$ target is itself a step curve, this tests the calibration, whereas the convergence in n below and the rank law (RQ4) test the theory. The CLIFF/IPA ratio follows $1 - e^{-\varepsilon}$ (Proposition 1) to within 0.01 at $q = 0.9$. For $q \le 0.99$, smaller ε makes CLIFF *cheaper* (1.23% at $\varepsilon = 0.25$ against 4.88% at $\varepsilon = 4$ for $q = 0.95$). At $q = 0.999$, the order reverses, as Remark 1 with the calibrated tolerances predicts: the protocol error, not negligible against $1 - q$ there, shrinks with ε faster than $1 - e^{-\varepsilon}$ grows. IPA, whose margin above $m^\star$ is at most $1 - q < \hat{\eta}_+$, never reaches a 0.9 hit rate on the p99.9 for $\varepsilon \le 1$ (below 0.65 at every tested budget up to 20%), where Remark 1 with $\hat{\eta}_+$ gives no budget ("∞"); at $q = 0.99$, $\varepsilon = 0.25$ ("†") it plateaus near the criterion (0.838–0.882 from 2% to 3.8%, 0.891–0.898 from 5% to 40%). Moreover, in Fig. 3(d), from $n = 10^5$ to 10^8, NBS's budget falls from 2.76% to 0.66%, approaching 0.63%; it is about the non-private threshold (1.00%) at $n = 10^6$ and drops below it from 3×10^6 on. For $n \ge 3 \times 10^5$, the prediction tracks NBS and BayeSS (measured up to 10^6) to within about 0.10 points (up to 0.23 at $n = 10^5$), and SGD-SN-log's to within 0.28 points at $n = 10^6$ and 3×10^6 (1.5 points too low at 3×10^5). We also re-estimated each 90%-hit budget β_{90} from 1000 fresh runs at each of 15 budgets (appendix): for $q \le 0.99$, $\hat{\beta}$ is within 2.4% of β_{90}, and $1.05\hat{\beta}$ reaches a one-sided 95% lower confidence bound of 0.9; for $q = 0.999$, within 11% and $1.15\hat{\beta}$. The analytic η_{NBS} ($\delta = 0.1$) certifies a larger budget (1.69% at $n = 10^6$, 0.74% at 10^8; dash-dotted), at which CLIFF returned exactly $m^\star$ in at least 99.5% of 200 runs for every $n \ge 3 \times 10^5$.

Answer to RQ1. For NBS and BayeSS, the cost of a $\pm 10\%$ hit depends mainly on rank, not value; SGD-SN-log's averaging lag adds a value-dependent cost. With tolerances calibrated on step curves, Remark 1 matches NBS's and BayeSS's budgets to within about 0.10 points ($n \ge 3 \times 10^5$), the budget converges to the asymptotic threshold, and an analytic η certifies NBS's budget. While η is small, stronger privacy makes the attack cheaper.

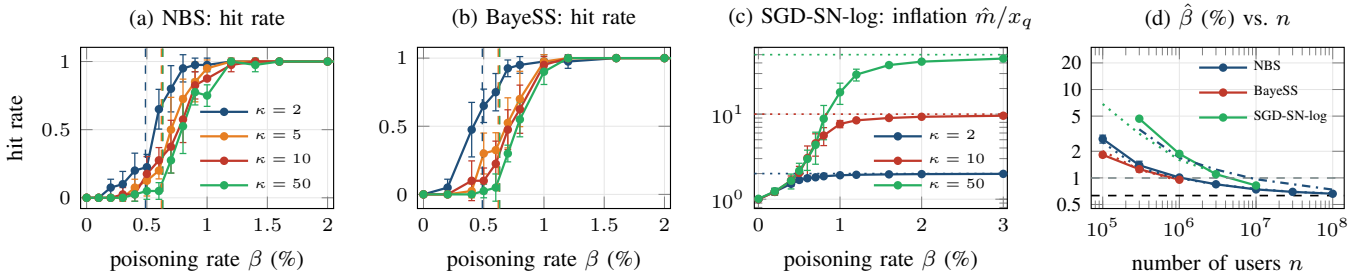


Fig. 3. RQ1 on CreditCard ($q = 0.99$, $\varepsilon = 1$, $n = 10^6$). (a,b) Hit rate of CLIFF versus β for $\kappa \in \{2, 5, 10, 50\}$ (40 runs per point); dashed: $\beta_0^\star(m^\star)$ (Theorem 1). (c) Inflation against SGD-SN-log; dotted: targets. (d) Empirical budget $\hat{\beta}$ for $\kappa = 10$ versus n; dotted: prediction of Remark 1 with step-calibrated $\hat{\eta}_\pm$; dash-dotted: budget certified for NBS ($\delta = 0.1$); gray: non-private threshold. SGD-SN-log at $n = 10^5$ misses a 0.9 hit rate within $[0, 6\%]$. Error bars: ±std.

TABLE II
RQ1: EMPIRICAL BUDGET $\hat{\beta}$ (%; NBS, $n = 10^7$, CREDITCARD) VERSUS THE REFERENCES "THM." AND "PRED."; >: ABOVE THE SEARCH RANGE; "THM.", "PRED.", ∞, †: SEE TEXT.

		$\varepsilon = 0.25$			$\varepsilon = 0.5$			$\varepsilon = 1$			$\varepsilon = 2$			$\varepsilon = 4$		
q	attack	emp.↓	thm.	pred.	emp.↓	thm.	pred.	emp.↓	thm.	pred.	emp.↓	thm.	pred.	emp.↓	thm.	pred.
0.9	CLIFF (OPA)	2.31 ± 0.01	2.15	2.31	3.95 ± 0.02	3.82	3.97	6.24 ± 0.01	6.14	6.28	8.47 ± 0.01	8.41	8.49	9.58 ± 0.00	9.55	9.61
	IPA	10.47 ± 0.11	9.73	10.48	10.04 ± 0.04	9.73	10.12	9.86 ± 0.01	9.73	9.94	9.79 ± 0.00	9.73	9.82	9.77 ± 0.01	9.73	9.79
0.95	CLIFF (OPA)	1.23 ± 0.01	1.09	1.26	2.07 ± 0.02	1.94	2.09	3.22 ± 0.01	3.11	3.23	4.34 ± 0.02	4.26	4.34	4.88 ± 0.00	4.83	4.88
	IPA	5.61 ± 0.07	4.92	5.72	5.25 ± 0.03	4.92	5.31	5.09 ± 0.01	4.92	5.11	5.00 ± 0.01	4.92	5.02	4.96 ± 0.01	4.92	4.97
0.99	CLIFF (OPA)	0.39 ± 0.01	0.22	0.39	0.53 ± 0.02	0.39	0.55	0.74 ± 0.00	0.63	0.75	0.93 ± 0.00	0.86	0.94	1.01 ± 0.00	0.98	1.01
	IPA	—†	1.00	∞	1.36 ± 0.03	1.00	1.39	1.18 ± 0.02	1.00	1.19	1.08 ± 0.01	1.00	1.09	1.02 ± 0.00	1.00	1.03
0.999	CLIFF (OPA)	0.21 ± 0.01	0.02	0.21	0.21 ± 0.01	0.04	0.19	0.19 ± 0.01	0.06	0.19	0.17 ± 0.01	0.09	0.16	0.12 ± 0.01	0.10	0.13
	IPA	>1.40	0.10	∞	>1.40	0.10	∞	>1.40	0.10	∞	0.22 ± 0.04	0.10	0.18	0.13 ± 0.00	0.10	0.13

TABLE III
RQ2: CLEAN RANK AND INFLATION, HIT RATE UNDER CLIFF, AND INFLATION AT 1% ($q = 0.99$, $\varepsilon = 1$, $m^\star = \min\{10x_q, B\}$). GREEN: ≥ 0.9; YELLOW: ≥ 0.5.

		clean		hit rate↑				infl.
	protocol	rank	infl.	β=0.5%	1%	1.5%	2%	@1%
CreditCard	NBS	0.992 ± 0.004	2.0 ± 3.9	0.05 ± 0.06	0.88 ± 0.11	1.00 ± 0.00	1.00 ± 0.00	9.4 ± 1.3
	BayeSS	0.991 ± 0.002	1.1 ± 0.2	0.30 ± 0.10	0.95 ± 0.10	1.00 ± 0.00	1.00 ± 0.00	9.8 ± 0.6
	SGD-SN-log	0.990 ± 0.001	1.0 ± 0.0	0.00 ± 0.00	0.00 ± 0.00	0.30 ± 0.13	0.98 ± 0.05	7.8 ± 0.6
	HH	0.982 ± 0.002	0.7 ± 0.1	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.63 ± 0.22	1.1 ± 0.1
	HH (T-CLIFF)	–	–	0.00 ± 0.00	0.08 ± 0.10	1.00 ± 0.00	1.00 ± 0.00	4.0 ± 2.6
Flights	NBS	0.991 ± 0.002	1.0 ± 0.1	0.15 ± 0.09	0.98 ± 0.05	1.00 ± 0.00	1.00 ± 0.00	9.9 ± 0.3
	BayeSS	0.992 ± 0.003	1.1 ± 0.1	0.15 ± 0.30	0.95 ± 0.10	1.00 ± 0.00	1.00 ± 0.00	9.9 ± 0.2
	SGD-SN-log	0.991 ± 0.001	1.0 ± 0.0	0.00 ± 0.00	0.03 ± 0.05	0.85 ± 0.05	1.00 ± 0.00	7.7 ± 0.8
	HH	0.989 ± 0.002	1.0 ± 0.0	0.00 ± 0.00	0.20 ± 0.19	1.00 ± 0.00	1.00 ± 0.00	3.4 ± 3.3
	HH (T-CLIFF)	–	–	0.00 ± 0.00	0.95 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	9.7 ± 1.3
Power	NBS	0.990 ± 0.003	1.0 ± 0.1	0.00 ± 0.00	0.85 ± 0.15	1.00 ± 0.00	1.00 ± 0.00	3.2 ± 0.5
	BayeSS	0.991 ± 0.002	1.0 ± 0.0	0.00 ± 0.00	0.85 ± 0.20	1.00 ± 0.00	1.00 ± 0.00	3.2 ± 0.4
	SGD-SN-log	0.991 ± 0.001	1.0 ± 0.0	0.00 ± 0.00	0.68 ± 0.13	1.00 ± 0.00	1.00 ± 0.00	3.1 ± 0.1
	HH	0.985 ± 0.002	0.9 ± 0.0	0.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	3.4 ± 0.0
	HH (T-CLIFF)	–	–	0.00 ± 0.00	0.85 ± 0.15	1.00 ± 0.00	1.00 ± 0.00	3.2 ± 0.4
Pareto(1.5)	NBS	0.991 ± 0.003	1.4 ± 1.9	0.18 ± 0.10	0.95 ± 0.06	1.00 ± 0.00	1.00 ± 0.00	9.7 ± 1.0
	BayeSS	0.991 ± 0.002	1.1 ± 0.2	0.00 ± 0.00	0.95 ± 0.10	1.00 ± 0.00	1.00 ± 0.00	9.9 ± 0.5
	SGD-SN-log	0.990 ± 0.001	1.0 ± 0.1	0.00 ± 0.00	0.00 ± 0.00	0.25 ± 0.22	1.00 ± 0.00	7.8 ± 0.5
	HH	0.981 ± 0.002	0.6 ± 0.1	0.00 ± 0.00	0.00 ± 0.00	0.23 ± 0.17	0.95 ± 0.06	1.1 ± 0.2
	HH (T-CLIFF)	–	–	0.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	3.5 ± 1.4

C. RQ2: Which protocols and data are vulnerable?

In Table III, NBS and BayeSS behave alike. At $\beta = 0.5\%$, below the target's own threshold $\beta_0^\star(m^\star) = 0.61$–$0.63\%$, the target is mostly missed (hit rate at most 0.30); 0.5% instead hits smaller targets (NBS, CreditCard, 10^8 users: $2x_q$ but not $3x_q$, with hit rates 1.00 and 0.00 and window thresholds 0.44% and 0.56%) or, with 10^7 users at $\varepsilon \leq 0.5$, where $\beta_0^\star(m^\star)$ falls to 0.22% and 0.39%, the tenfold one (hit rates 1.00 and 0.88 at $\varepsilon = 0.25$ and 0.5). At 1%, they hit the target in 85–98% of runs, and the mean achieved inflation is 9.4–9.9× (3.2× on Power, whose target is capped at $3.4x_q$); from 1.5% on, they hit it in every run. Theorem 1 with Remark 1 certifies exact landing for NBS ($\delta = 0.1$) from 1.69% on. Empirically, at 1% (10,000 runs), NBS hits the target in 89.8% of runs (95% interval $[89.1, 90.3]$%) and lands exactly in 39.3%; at 1.5%, it lands exactly in every run, whereas BayeSS does so in 55%–85% of hits and SGD-SN-log never. The default $n = 10^6$ is demanding for the protocols themselves: in 10,000 clean runs on CreditCard, NBS outputs at least $2x_q$ in 5.0% of runs (none at $n = 10^7$). SGD-SN-log needs more at $n = 10^6$ (RQ1): at 1% it is inflated 7.7–7.8× on the datasets with headroom and hits the target in at most 68% of runs (Power), at 2% in at least 97.5%; with $n = 10^7$, 1% suffices on all four datasets. Line 3 of Algorithm 1 offsets this lag: the attacker simulates SGD-SN-log with the step at $m^\star$ on a synthetic population with its p99 at $m^\star/10$, measures where the average lands, $\tilde{m} < m^\star$, and places the step ahead of the target, at $\lambda m^\star$ with $\lambda = m^\star/\tilde{m}$ (1.12–1.28), as one leads a moving target; this heuristic assumes that the lag scales with the step position and needs a guess of x_q. With the shifted step, on CreditCard, Flights, and Pareto, CLIFF hits in 0.83–0.98 of runs at 1.2% and 0.95–0.98 at 1.4%, plain CLIFF in at most 0.53 (Power: plain CLIFF at 1.5%).

HH's consistency step partly absorbs generic CLIFF (§IV-E):

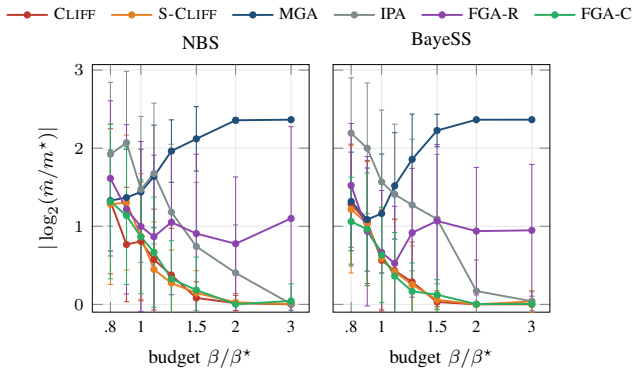


Fig. 4. RQ3: landing error $|\log_2(\hat{m}/m^\star)|$ versus budget relative to $\beta^\star$ (CreditCard, $n = 10^6$, $\kappa = 10$). CLIFF is insensitive to over-provisioning; MGA overshoots to the boundary; IPA needs more; FGA-R hits in at most half of the runs; FGA-C comes close to CLIFF.

in a separate sweep, it needed 1–3% for a hit rate of 0.95 (3% on CreditCard) (appendix). T-CLIFF injects a consistent shift and lands exactly on the target in every run at 1.5% on all four datasets; at 1% its hit rate is 0.00–0.95. Its cost is set by HH's error, close to $1 - q$ at $n = 10^7$ (clean rank 0.981–0.989); as it shrinks, the cost decreases toward the asymptotic 0.46% of Proposition 3: with 10^8 users, T-CLIFF hits the target with 0.75% (hit rate at least 0.95), whereas generic CLIFF at 1% still fails on CreditCard (0.00; it hits the other three). RQA cannot be steered and floods the p99 toward the domain edge already at 0.5% (CreditCard, Flights, Pareto).

Answer to RQ2. Every evaluated protocol and dataset is vulnerable: at most 1.5% fake users place the p99 within 10% of ten times its value in at least 95% of runs, and NBS lands on it exactly in every run. Only SGD-SN-log needs more (2% at $n = 10^6$), and HH needs the tree-consistent T-CLIFF.

D. RQ3: What does the attacker need?

Fig. 4 sweeps the budget from $0.8\beta^\star$ to $3\beta^\star$ ($\beta^\star = 0.63\%$ for $\kappa = 10$). CLIFF's landing error falls with the budget and is near zero from about $2\beta^\star$ (hit rate 0.98 for NBS, 1.00 for BayeSS); over-provisioning never hurts. S-CLIFF, which answers honestly outside $[x_q/2, m^\star)$, stays within 0.1 of CLIFF's hit rate from $\beta^\star$ on (20–40 runs per cell). MGA [35] has no step at $m^\star$: it lands at the $\frac{q+\beta c_\varepsilon}{1-\beta}$-quantile of F (or at B), which moves steeply with β in the flat tail, and hits the target in at most 10% of runs (SGD-SN-log: 15%) on the grid of Fig. 4. It reaches $0.9m^\star$ at least as often as CLIFF, enough for a one-sided goal, but from $1.5\beta^\star$ on (NBS, BayeSS) lands in the top half of the domain, far above $m^\star$, in at least 0.85 of runs, whereas CLIFF never does and S-CLIFF does in at most 0.05 at any budget (appendix); hence, it cannot set a chosen value (BayeSS, $1.5\beta^\star$: 47× instead of 10×). IPA, the attacker facing verifiable randomization [43], needs about $3\beta^\star$ for a 0.9 hit rate under NBS and BayeSS (SGD-SN-log: more), consistent with the factor 1.58 of Proposition 1 plus the protocol error. FGA-R [37] knows the honest CDF yet hits the target in at most 0.50 of runs at any budget: its target, the honest data with the mass of $[x_q, m^\star)$ moved to $m^\star$, leaves the curve on $[x_q, m^\star)$ below q by at most the atom at x_q ($<10^{-5}$ on CreditCard), so the protocol stops anywhere there (4.0–7.8× on average; SGD-SN-log: 1.4–1.6×). FGA-C, which targets CLIFF's curve clipped to $[0, 1]$ and so keeps margins of up to q below $m^\star$ and $1 - q$ above it, stays within 0.20 of CLIFF's hit rate at every budget and protocol (20–40 runs per cell). With 2,000 NBS runs per cell, it is within 2 points of CLIFF up to $1.25\beta^\star$, but from $1.5\beta^\star$ on, it is significantly less reliable (0.95 against 0.99 at $2\beta^\star$; Fisher's exact test, $p < 10^{-12}$). Thus, a cliff needs no knowledge of F, although knowing $F(m^\star - 1)$ lowers the budget for targets of rank well below 1.

TABLE IV
RQ3: BUDGET PLANNED ON SYNTHETIC DATA (%) AND HIT RATE ON THE REAL DATA AT 1× AND 1.25× THE PLAN.

	CreditCard			Flights			Power		
		hit rate↑			hit rate↑			hit rate↑	
protocol (n)	plan	1×	1.25×	plan	1×	1.25×	plan	1×	1.25×
NBS (10^6)	1.14	1.00 ± 0.00	1.00 ± 0.00	0.86	0.90 ± 0.09	1.00 ± 0.00	1.15	0.98 ± 0.05	1.00 ± 0.00
BayeSS (10^6)	0.97	0.95 ± 0.06	1.00 ± 0.00	0.97	0.95 ± 0.06	1.00 ± 0.00	1.00	0.85 ± 0.12	1.00 ± 0.00
SGD-SN-log (10^6)	1.80	0.83 ± 0.15	1.00 ± 0.00	1.71	0.93 ± 0.06	1.00 ± 0.00	1.80	1.00 ± 0.00	1.00 ± 0.00
SGD-SN-log (10^7)	0.77	0.55 ± 0.23	1.00 ± 0.00	0.77	0.50 ± 0.16	0.98 ± 0.05	0.77	1.00 ± 0.00	1.00 ± 0.00
HH (T-CLIFF) (10^7)	1.27	0.80 ± 0.06	1.00 ± 0.00	0.89	0.98 ± 0.05	1.00 ± 0.00	0.98	0.90 ± 0.12	1.00 ± 0.00

CLIFF itself needs no data: for $\kappa = 10$, $F(m^\star) \approx 1$, so $\beta^\star$ is the universal bound of Corollary 1. Table IV runs the calibration of Algorithm 1 (lines 1–6, without the shift of line 3) with a classic (type-I) Pareto(1.5) surrogate and $1 - \delta = 0.95$, and attacks the real data with the calibrated budget, without ($\mu = 0$) and with ($\mu = 0.25$) a margin. Without a margin, the plan transfers only roughly: at the planned budget (planned for 0.95), the hit rate on the real data is at least 0.9 in 10 of the 15 settings (median 0.93). The misses are mostly SGD-SN-log (as low as 0.50), whose averaged estimate lags just below the window (RQ1). With the margin $\mu = 0.25$, the hit rate is at least 0.975 in every setting, for every protocol, including HH (plan 0.89–1.27%). The transfer is not guaranteed: if the CDF stays at q above x_q (mass 0.99 at $x_q = 100$ and 0.01 at B), the margin above $m^\star$ shrinks to $\beta(1 + c_\varepsilon - q)$, and NBS with the plan of 1.19% hits $m^\star = 10x_q$ in only 0.61 of runs (0.83 with $\mu = 0.25$); the budget that Remark 1 certifies with η_{NBS} and worst-case ranks, 2.83%, covers it (exact in 99.95% of 2000 runs). The attacker need not know x_q for the rate: planned for $\hat{\kappa} = 1000$, i.e., for $F(m^\star - 1) \approx 1$, CLIFF with $\mu = 0.25$ hit the target in every run in all 15 settings, at up to 2.08 times the budget planned with the true κ (SGD-SN-log), whereas, in a further run, $\hat{\kappa} = 2$ under-provisioned (appendix). The shifted step of line 3 does need $\hat{\kappa}$: at 1.4%, $\hat{\kappa} \in \{5, 10, 20\}$ (true 10) gives hit rates of 0.70–1.00, but $\hat{\kappa} = 1000$ only 0.00–0.35 (CreditCard, Flights, Pareto). S-CLIFF also needs a rough lower bound on x_q.

TABLE V
RQ4: BUDGET (%) TO HIT $\min\{10x_q, B\}$ BY QUANTILE LEVEL q (NBS, $n = 10^7$, $\varepsilon = 1$).

	$\hat{\beta}$ (%)↓ at quantile level q					
dataset	0.5	0.75	0.9	0.95	0.99	0.999
CreditCard	26.91 ± 0.02	14.81 ± 0.01	6.24 ± 0.01	3.22 ± 0.01	0.73 ± 0.02	0.20 ± 0.02
thm./pred.	26.78/26.88	14.73/14.89	6.14/6.28	3.11/3.23	0.63/0.75	0.06/0.19
Flights	31.73 ± 0.00	15.92 ± 0.01	6.42 ± 0.01	3.28 ± 0.01	0.70 ± 0.00	0.17 ± 0.01
thm./pred.	31.60/31.73	15.80/15.93	6.32/6.41	3.16/3.27	0.63/0.71	0.06/0.17
Power	31.44 ± 0.01	15.92 ± 0.01	6.46 ± 0.01	3.29 ± 0.00	0.75 ± 0.01	0.19 ± 0.01
thm./pred.	31.37/31.50	15.80/15.93	6.32/6.45	3.16/3.29	0.63/0.76	0.06/0.19
Pareto(1.5)	28.83 ± 0.01	14.92 ± 0.01	6.13 ± 0.02	3.14 ± 0.01	0.71 ± 0.01	0.22 ± 0.01
thm./pred.	28.75/28.92	14.85/15.00	6.02/6.16	3.03/3.17	0.61/0.73	0.06/0.21
$(1-q)(1-e^{-\varepsilon})$	31.61	15.80	6.32	3.16	0.63	0.06

Answer to RQ3. The attacker needs no real data: Algorithm 1 calibrates the budget on a synthetic population, and with a 25% margin, the plan hit the target on every evaluated dataset, unless the CDF stays at q above the target. The step is what makes the attack precise: the baselines overshoot, stop short, or need more fake users, and FGA-C, the closest, needs the honest CDF F and is less reliable at large budgets.

E. RQ4: When does the attack become expensive?

Quantile level. Table V varies q on all four datasets. In every cell, the empirical budget exceeds the window threshold ("thm.") by 0.07–0.16 percentage points, a gap due to the protocol error at $n = 10^7$, and the finite-sample prediction matches it to within 0.09 points. Moving the median tenfold costs 27–32% of users, the p99 0.70–0.75% (asymptotic threshold at most 0.63%), and the p99.9 0.17–0.22% (at most 0.06%), where the error dominates. Only central quantiles are expensive: up to the protocol error, the budget scales with $1-q$ (Observation 1).

Tail shape and headroom. We also placed the cliff at the domain edge on all seven synthetic distributions (appendix): below the universal budget, the p99 moves only to the $\frac{q+\beta c_\varepsilon}{1-\beta}$-quantile of F as $\eta \to 0$, so heavier tails give more (1.65× in geometric mean for Pareto(1.1) at $\beta = 0.25\%$, predicted 1.59×), and at 1%, every distribution reaches B in every run. The damage is set by the headroom B/x_q.

Direction. We also measured deflation (appendix): moving the q-th quantile down to x_{q-d} costs more than moving it up to x_{q+d} (1.31% against 0.49% for $q = 0.99$ and $d = 0.005$), and for $q \in \{0.95, 0.99\}$, the ratio, 2.3–2.7, lies within 0.04 of the ratio of Proposition 2's bounds in all but two cells (up to 0.26, from the protocol error). Hence, hiding a regression costs more than evading a threshold.

Answer to RQ4. The attack becomes expensive for central quantiles, and deflation costs up to e^ε times more than inflation. For upper quantiles, the only limit is the headroom of the domain, which bounds the damage but not the cost.

F. RQ5: Can the defender detect or prevent it?

Detection. In Table VI ($\varepsilon = 1$; outputs $\hat{m}$ of NBS), an extra cohort of 5% or 20% of n users answers audit queries, and each audit flags a deviation beyond 3 standard errors.

TABLE VI
RQ5: FRACTION OF 200 RUNS FLAGGED BY THREE AUDITS (NBS, CREDITCARD, $\beta = 1\%$). CLEAN: FALSE-POSITIVE RATE.

		FPR↓	detection rate↑				
n	audit	clean	CLIFF	S-CLIFF	IPA	MGA	LOW
range audit							
10^6	5%	0.025	0.025	0.015	0.010	0.005	0.005
10^7	5%	0.005	0.070	0.015	0.015	0.000	0.000
10^7	20%	0.015	0.280	0.005	0.005	0.000	0.000
10^8	20%	0.000	1.000	0.005	0.000	0.000	0.000
jump audit							
10^6	5%	0.000	0.010	0.000	0.000	0.005	0.005
10^7	5%	0.000	0.665	0.080	0.000	0.700	0.000
10^7	20%	0.000	1.000	0.420	0.000	1.000	0.000
10^8	20%	0.000	1.000	1.000	0.000	1.000	0.000
boundary audit ($t \in \{0, B\}$)							
10^6	5%	0.000	0.075	0.000	0.005	0.045	0.000
10^7	5%	0.005	0.730	0.000	0.000	0.810	0.000
10^7	20%	0.000	1.000	0.000	0.000	1.000	0.005
10^8	20%	0.010	1.000	0.005	0.000	1.000	0.000

The *range audit* flags estimates outside $[0, 1]$ at 31 log-spaced thresholds, which CLIFF exceeds beyond $m^\star$ by only $\beta c_\varepsilon \approx 0.006$; it is flagged in 2.5% (the clean rate), 28%, and 100% of runs at $n = 10^6$, 10^7, 10^8 (20% cohort from 10^7). S-CLIFF's curve stays in $[0, 1]$ whenever $F(L) > \beta c_\varepsilon/(1-\beta)$, so no range audit can see it. Tail saturation also helps the defender: an honest population has mass below $1 - q + \eta$ above an accurate estimate $\hat{m}$, so the mass of $[\hat{m}, 1.1\hat{m}]$, estimated at $\hat{m}-1$ and $1.1\hat{m}$, is below $1-q+\eta+\Pr[x = \hat{m}]$ (*jump audit*). On CreditCard, every observed $\hat{m}$ lies above 750 EUR, where atoms are below 4.6×10^{-4}, so we drop the last term (the audit thus uses knowledge of the data and the empirical $\hat{\eta}_-$, so its false-positive rate is measured, not certified); CLIFF's step adds $\beta(1 + 2c_\varepsilon) = 2.2\%$. The jump audit never flagged clean data and has no power at $n = 10^6$. At $n = 10^7$, it detects CLIFF and MGA (whose step sits at B) in 0.67 and 0.70 of runs with a 5% cohort and in every run with 20%; S-CLIFF (step $\beta(1+c_\varepsilon)$) less often (Table VI). A *boundary* audit at $t \in \{0, B\}$ (half the cohort each), where honest answers debias to means 0 and 1, sees CLIFF's shift βc_ε in 73% of runs at $n = 10^7$ with a 5% cohort (false positives at most 1%). An adaptive attacker evades all three. LOW-CLIFF answers honestly below $x_q/2$ and from $2m^\star$ on, "no" up to $m^\star$, and on $[m^\star, 2m^\star)$ "yes" only as often as needed to reach $q + \hat{\eta}_+$ (debiased value v). No audit flagged it in more than 1 of 200 runs, yet it hit the target in 0.850, 0.895, and 0.950 of runs at $n = 10^6$, 10^7, and 10^8. At $n = 10^8$, its jump statistic averages 0.0050 against 0.0015 on clean data, both below $1 - q + \hat{\eta}_-$; a test calibrated on clean history might catch it. Per-user report detectors [35], [44] see only valid bits. On HH, a non-negativity audit of the raw node estimates flags T-CLIFF in every run on Flights, Power, and Pareto (1.5%, $n = 10^7$) but in only 0.13 on CreditCard; we also ran the spread variant, which it flags in at most 0.05 (clean: 0.010–0.020) (appendix). Recovery and robust aggregation [45], [47], [48] do not target adaptive threshold protocols (on HH, we leave them to future work).

TABLE VII
RQ5: CTQ AT $q^\dagger$ ON CREDITCARD ($n = 10^6$). NBS†: CERTIFIED η_{NBS}; OTHER BLOCKS: ORACLE η. ARROWS: DEFENDER'S VIEW.

		NBS† (certified)		NBS		BayeSS		SGD-SN-log		BayeSS
ε	$\bar{\beta}$	clean rank↑	safe@$\bar{\beta}$↑	clean rank↑	safe@$\bar{\beta}$↑	clean rank↑	safe@$\bar{\beta}$↑	clean rank↑	safe@$\bar{\beta}$↑	infl.@$2\bar{\beta}$↓
0.5	0.5%	0.946 ± 0.006	1.00 ± 0.00	0.954 ± 0.004	1.00 ± 0.00	0.960 ± 0.005	1.00 ± 0.00	0.969 ± 0.002	1.00 ± 0.00	0.90 ± 0.18
	1%	0.932 ± 0.005	1.00 ± 0.00	0.943 ± 0.005	0.98 ± 0.05	0.947 ± 0.005	1.00 ± 0.00	0.957 ± 0.002	1.00 ± 0.00	4.40 ± 3.15
	2%	0.907 ± 0.005	1.00 ± 0.00	0.918 ± 0.005	1.00 ± 0.00	0.922 ± 0.005	1.00 ± 0.00	0.931 ± 0.002	1.00 ± 0.00	10.00 ± 0.00
1	0.5%	0.965 ± 0.004	1.00 ± 0.00	0.965 ± 0.003	1.00 ± 0.00	0.970 ± 0.002	1.00 ± 0.00	0.978 ± 0.001	1.00 ± 0.00	0.82 ± 0.09
	1%	0.958 ± 0.003	1.00 ± 0.00	0.957 ± 0.003	1.00 ± 0.00	0.962 ± 0.002	1.00 ± 0.00	0.970 ± 0.001	1.00 ± 0.00	1.53 ± 0.35
	2%	0.941 ± 0.003	1.00 ± 0.00	0.942 ± 0.003	1.00 ± 0.00	0.946 ± 0.002	1.00 ± 0.00	0.954 ± 0.001	1.00 ± 0.00	10.00 ± 0.00
2	0.5%	0.974 ± 0.001	1.00 ± 0.00	0.979 ± 0.001	1.00 ± 0.00	0.978 ± 0.001	1.00 ± 0.00	0.982 ± 0.001	1.00 ± 0.00	0.98 ± 0.07
	1%	0.969 ± 0.001	1.00 ± 0.00	0.974 ± 0.001	1.00 ± 0.00	0.972 ± 0.001	1.00 ± 0.00	0.977 ± 0.001	1.00 ± 0.00	1.72 ± 0.52
	2%	0.957 ± 0.001	1.00 ± 0.00	0.962 ± 0.001	0.98 ± 0.05	0.961 ± 0.002	1.00 ± 0.00	0.965 ± 0.000	1.00 ± 0.00	10.00 ± 0.00

Change monitoring. Operators also watch how a threshold moves. A monitor that flags a period whose estimate exceeds twice the previous one never fired in 8 clean periods of NBS ($n = 10^7$, CreditCard, 2,000 runs) but flagged MGA and a one-shot CLIFF to $10x_q$ ($\beta = 1\%$) in every run. Retargeted each period to 1.6 times the previous estimate, however, CLIFF reached $10x_q$ within $\pm 10\%$ in five periods in 91.7% of runs ($[90.4, 92.8]\%$) without an alarm: only a precise attack stays below such a monitor. At $n = 10^6$, the monitor itself is unusable (clean false alarms in 5.6% of periods).

Certified defense. Table VII evaluates CTQ. With the analytic η_{NBS} (first block), Theorem 3 certifies NBS: CTQ kept the output at or below the true p99 in every run at $\beta = \bar{\beta}$ in all nine settings, at a clean rank of 0.958 for $\varepsilon = 1$ and $\bar{\beta} = 1\%$ (0.969 with 10^7 users). An empirical margin must still cover the protocol error: with a fixed $\eta = 0.001$, BayeSS stayed at or below x_q in as few as 5% of runs. The other blocks set η to four standard deviations of the clean output rank on the evaluated data (at $\varepsilon = 1$: NBS 0.017, BayeSS 0.013, SGD-SN-log 0.004), an *oracle* margin a collector would have to estimate and that Theorem 3 does not cover (except NBS at $\varepsilon = 1$, where 0.0170 happens to exceed $\eta_{\mathrm{NBS}} = 0.0167$). With it, CTQ kept the output at or below the true p99 in at least 0.975 of runs in all 27 settings (BayeSS: 20 runs each), at about the same price (NBS: 0.957). Adaptive targets ($x_q + 1$ to $2x_q$) and the non-monotone S-CLIFF (45 settings; $\varepsilon = 1$, $n = 10^6$, CreditCard) never pushed the output above x_q. Outside the certificate ($\beta = 2\bar{\beta}$), protection degrades (BayeSS, $\varepsilon = 1$: $1.5\times$ at $\beta = 2\%$ for $\bar{\beta} = 1\%$, the full $10\times$ at 4% for $\bar{\beta} = 2\%$). The price grows with $\bar{\beta}$, c_ε, η; the first two are unavoidable (Proposition 4). In value it is large: NBS's certified clean output on CreditCard is $0.41x_q$.

Answer to RQ5. On NBS, the three audits miss CLIFF or need large cohorts, and a lowered step evades them. A change monitor catches untargeted and one-shot pushes but not a ramped CLIFF. CTQ with an analytic margin provably prevents inflation within its budget, at a known rank shift; oracle margins kept at least 0.975 of runs safe in RQ5.

G. RQ6: What are the downstream consequences?

(a) Evading a review threshold. Payment services that do not share their customers' amounts set a common review threshold at the p99 of these amounts, collected under LDP (true value 1,018 EUR); the attacker wants 5,000 transactions, log-uniform on $(x_{0.99}, 10x_{0.99}]$, to pass. With 1% fake users, the BayeSS threshold rises from about 1,208 to 10,210 EUR, and 99.8% evade review (7% without attack). With CTQ ($\bar{\beta} = 1\%$, oracle margins), at most 0.02% evade review up to $\beta = 1\%$ for any protocol, at the price of a lower clean threshold (BayeSS: 452 EUR, reviewing 3.78% of legitimate traffic).

(b) Confidence hijacking. SGD-SN-log's self-normalized interval [11] covers the true p99 in 0.932 (discrete data) and 0.945 (buckets spread uniformly) of 1000 clean runs. On the continuous data, coverage falls to 0.04 at $\beta = 0.5\%$ and to 0 from 1% on, where it covers $m^\star$ in every run, around a forged $7.7 \times x_q$; we also ran the raw-scale original (clean: 0.883), which likewise stops covering the true p99 (appendix).

(c) False alarms. We also simulated an SLO alarm on Flights that fires when the estimated p99 delay exceeds twice the true one (274 min): CLIFF at 1%, targeting three times the p99, fires it in every run (clean: at most 5%) (appendix).

Answer to RQ6. A pinned threshold hides transactions, triggers false alarms, and forges a confidence interval; an untargeted push serves these one-sided goals too, but it sets no chosen value and trips a change monitor (RQ5).

VII. DISCUSSION AND CONCLUSION

Up to the protocol's error, moving an LDP tail quantile costs rank, not value: rank-space accuracy, which suits *utility* [12], hides a domain-scale value error from *security*. Above $(1 - q)(1 - e^{-\varepsilon})$ fake users, CLIFF pins any target above x_q once the protocol's error is small, asymptotically optimally for tail targets; CTQ certifies against it at a price in rank. Adaptive clipping in differentially private federated learning uses the same threshold bits [2]; an inflated clipping norm admits larger poisoned updates, at a cost set by the targeted quantile.

Limitations. Distribution estimators [23], non-linear post-processing [31], and attackers who control when their accounts answer remain open. Only NBS's CTQ is certified, given $\bar{\beta}$, which fake-account detection [44], [46] could supply.

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APPENDIX A
FULL PROOFS

Throughout, $\rho = \rho_\varepsilon = 1/(1+e^\varepsilon)$, $c_\varepsilon = \rho/(1-2\rho) = 1/(e^\varepsilon - 1)$, and $1-2\rho = (e^\varepsilon-1)/(e^\varepsilon+1)$. For a fake-user strategy s and poisoning rate β, a query at threshold t is answered by an honest user with probability $1-\beta$ and by a fake user with probability β, independently of everything else (contamination model, §III). Hence the answer is a Bernoulli bit with mean $\pi_\beta(t) = (1-\beta)\pi(t) + \beta s(t)$, and the debiased curve is

$$G_\beta(t) = \frac{\pi_\beta(t)-\rho}{1-2\rho} = (1-\beta)\frac{\pi(t)-\rho}{1-2\rho} + \beta\frac{s(t)-\rho}{1-2\rho}$$
$$= (1-\beta)F(t) + \beta g(t).$$

A. Proof of Lemma 1

The map $s \mapsto g = (s-\rho)/(1-2\rho)$ is increasing. For $s \in [0,1]$ it ranges over $[-\rho/(1-2\rho), (1-\rho)/(1-2\rho)] = [-c_\varepsilon, 1+c_\varepsilon]$, because $(1-\rho)/(1-2\rho) = 1 + \rho/(1-2\rho)$. Substituting the endpoints into $G_\beta(t) = (1-\beta)F(t) + \beta g(t)$ gives the OPA envelope. Under IPA, a fake user holding value v answers $\mathrm{RR}_\varepsilon(\mathbf{1}[v \le t])$, so $s(t) \in \{\rho, 1-\rho\}$ for a fixed v. A mixture over values keeps $s(t) \in [\rho, 1-\rho]$, hence $g(t) \in [0,1]$, which gives the IPA envelope. □

B. Proof of Lemma 2

The randomizer has two inputs, $\mathbf{1}[x \le t] = 1$ and 0, and two outputs. ε-LDP for output 1 gives $a \le e^\varepsilon b$, i.e., $a - b \le (e^\varepsilon - 1)b$, so a reported 0 counts $-b/(a-b) \le -1/(e^\varepsilon - 1) = -c_\varepsilon$. ε-LDP for output 0 gives $1-b \le e^\varepsilon(1-a)$, i.e., $a - b = (1-b) - (1-a) \le (e^\varepsilon - 1)(1-a)$, so a reported 1 counts $(1-b)/(a-b) = 1 + (1-a)/(a-b) \ge 1 + c_\varepsilon$. Both bounds are tight if and only if $a = e^\varepsilon b$ and $1 - b = e^\varepsilon(1-a)$, whose unique solution is $b = 1/(1+e^\varepsilon) = \rho_\varepsilon$ and $a = 1 - \rho_\varepsilon$, i.e., RR_ε. Consequently, a fake user can realize every debiased contribution in $[-c_\varepsilon, 1+c_\varepsilon]$ under any binary ε-LDP randomizer, by answering 1 with probability $b + (a-b)g$, and the envelope of Lemma 1 only widens. For a fixed pair (a,b) and with Definition 1 applied to the debiased answers, the proofs of Theorem 1 and Corollary 1 use only that a fake user can contribute $-c_\varepsilon$ below $m^\star$ and $1+c_\varepsilon$ from $m^\star$ on, so they hold verbatim. □

C. Proof of Theorem 1

Under $\textsc{Cliff}(m^\star)$, $s(t) = \mathbf{1}[t \ge m^\star]$, so $g(t) = -c_\varepsilon$ for $t < m^\star$ and $g(t) = 1 + c_\varepsilon$ for $t \ge m^\star$, which yields (4).

Monotonicity. G_β is non-decreasing on $\{0, \dots, m^\star - 1\}$ and on $\{m^\star, \dots, B\}$ because F is. At the junction, $G_\beta(m^\star) - G_\beta(m^\star - 1) = (1-\beta)(F(m^\star) - F(m^\star - 1)) + \beta(1 + 2c_\varepsilon) > 0$. Hence π_β is a non-decreasing curve with values in $[0,1]$, and Definition 1 applies to it.

Below the target. For $t < m^\star$, monotonicity of F gives $G_\beta(t) \le (1-\beta)F(m^\star - 1) - \beta c_\varepsilon$. Condition (5) is equivalent to $\beta(F(m^\star - 1) + c_\varepsilon) \ge F(m^\star - 1) - q + \eta$, i.e., $(1-\beta)F(m^\star - 1) - \beta c_\varepsilon \le q - \eta$. Hence $G_\beta(t) \le q - \eta$ for all $t < m^\star$.

At and above the target. If $m^\star < B$, then $F(m^\star) \ge q + \eta$ and, for $t \ge m^\star$, $G_\beta(t) = F(t) + \beta(1 + c_\varepsilon - F(t)) \ge F(t) \ge F(m^\star) \ge q + \eta$, since $1 + c_\varepsilon - F(t) \ge c_\varepsilon > 0$. If $m^\star = B$, no output lies above the target.

Conclusion. With probability at least $1 - \delta$, the output satisfies conditions (a) and (b) of Definition 1. If $\hat{m} < m^\star$, then $\hat{m} < B$, so (b) requires $G_\beta(\hat{m}) > q - \eta$, whereas $G_\beta(\hat{m}) \le q - \eta$, a contradiction. If $\hat{m} > m^\star$, then $\hat{m} > 1$, so (a) requires $G_\beta(\hat{m} - 1) < q + \eta$, whereas $\hat{m} - 1 \ge m^\star$ gives $G_\beta(\hat{m} - 1) \ge q + \eta$, a contradiction. Hence $\hat{m} = m^\star$. □

D. Proof of Corollary 1

Let $h(F) = (F - q + \eta)/(F + c_\varepsilon)$. Then $h'(F) = (c_\varepsilon + q - \eta)/(F + c_\varepsilon)^2 > 0$ because $\eta < q + c_\varepsilon$ by assumption. So h is increasing and $\beta^\star_\eta(m^\star) = h(F(m^\star - 1)) \le h(1) = (1 - q + \eta)/(1 + c_\varepsilon)$. Finally, $1 + c_\varepsilon = e^\varepsilon/(e^\varepsilon - 1)$, so $1/(1 + c_\varepsilon) = 1 - e^{-\varepsilon}$. □

E. Justification of Remark 1

In the proof of Theorem 1, the step "at and above the target" only uses $G_\beta(t) \ge q + \eta$ for $t \ge m^\star$. Under CLIFF, $G_\beta(t) = F(t) + \beta(1 + c_\varepsilon - F(t))$ is non-decreasing in t and increasing in β, so this holds for all $t \ge m^\star$ once it holds at $m^\star$, i.e., once $\beta \ge (q + \eta - F(m^\star))/(1 + c_\varepsilon - F(m^\star))$. Under IPA, $g \le 1$ gives $G_\beta \le (1-\beta)F + \beta \le 1$. If $\eta > 1 - q$, then $G_\beta(\hat{m} - 1) < q + \eta$ for every output, so condition (a) of Definition 1 excludes no output above the target, and the argument cannot pin $m^\star < B$. For $m^\star = B$, no output lies above the target, and condition (b) alone forces $\hat{m} = B$.

Window bound. Let $1 < \ell \le m^\star \le u \le B$ and let fake users follow $\textsc{Cliff}(m^\star)$, whose curve (4) is non-decreasing, so Definition 1 applies. For $t \le \ell - 1 < m^\star$, $G_\beta(t) \le (1-\beta)F(\ell - 1) - \beta c_\varepsilon \le q - \eta$ by the first condition; an output $\hat{m} \le \ell - 1$ is below B, so (b) would require $G_\beta(\hat{m}) > q - \eta$, a contradiction. If $u < B$, the second condition gives $G_\beta(u) = (1-\beta)F(u) + \beta(1 + c_\varepsilon) \ge q + \eta$; an output $\hat{m} > u$ is above 1, so (a) would require $G_\beta(\hat{m} - 1) < q + \eta$, whereas $G_\beta(\hat{m} - 1) \ge G_\beta(u)$. Hence $\ell \le \hat{m} \le u$ with probability at least $1 - \delta$; for $u = B$ the upper end holds trivially. □

Search path. The actual finite-n budget also depends on the search path, not only on the ranks: for NBS on $B = 8$ with all honest values at 1 ($q = 0.99$, $\varepsilon = 1$, $n = 3 \times 10^5$), targets 2 and 7 have equal ranks, but 7 needs two thin-margin decisions instead of one and a 90%-hit budget of 0.95% instead of 0.88% (exact computation).

F. Proof of Theorem 2

Fix a time-invariant strategy s for which π_β is non-decreasing, so that Definition 1 applies; this includes every non-decreasing s, since π is non-decreasing. If Π is robustly accurate, s may be any time-invariant strategy. By Lemma 1 and monotonicity of F, for every $t \ge m^\star - 1$,

$$G_\beta(t) \ge (1-\beta)F(t) - \beta c_\varepsilon \ge (1-\beta)F(m^\star - 1) - \beta c_\varepsilon > q + \eta,$$

where the last inequality is the assumption $\beta < (F(m^\star - 1) - q - \eta)/(F(m^\star - 1) + c_\varepsilon)$ rearranged. The assumption also implies $F(m^\star - 1) > q$, hence $m^\star \ge 2$. With probability

at least $1-\delta$, condition (a) holds: either $\hat{m}=1<m^\star$, or $G_\beta(\hat{m}-1)<q+\eta$, hence $\hat{m}-1<m^\star-1$. In both cases $\hat{m}<m^\star$. Only pointwise bounds on G_β were used. □

G. Proof of Proposition 1

Under IPA, the fake-user contribution satisfies $g(t)\in[0,1]$ (Lemma 1). *Sufficiency:* fake users holding $m^\star$ answer $\mathrm{RR}_\varepsilon(\mathbf{1}[m^\star\le t])$, so $g(t)=\mathbf{1}[t\ge m^\star]$ exactly. Then $G_\beta(t)=(1-\beta)F(t)\le(1-\beta)F(m^\star-1)$ for $t<m^\star$, and $G_\beta(t)=(1-\beta)F(t)+\beta\ge F(t)\ge F(m^\star)\ge q+\eta$ for $t\ge m^\star$. The argument of Theorem 1 goes through with c_ε replaced by 0 whenever $(1-\beta)F(m^\star-1)\le q-\eta$, i.e., $\beta\ge(F(m^\star-1)-q+\eta)/F(m^\star-1)$. *Necessity:* for any IPA strategy that keeps the curve non-decreasing, if $\beta<(F(m^\star-1)-q-\eta)/F(m^\star-1)$, then for every $t\ge m^\star-1$, $G_\beta(t)\ge(1-\beta)F(m^\star-1)>q+\eta$, and condition (a) forces $\hat{m}<m^\star$ as in the proof of Theorem 2. The non-private statement is the case $\varepsilon\to\infty$ (no randomization, $\rho=0$) with an exact quantile ($\eta=0$): contamination by a β fraction of users holding $m^\star$ gives the mixture CDF $(1-\beta)F+\beta\mathbf{1}[\cdot\ge m^\star]$, whose q-quantile equals $m^\star$ exactly when $(1-\beta)F(m^\star-1)<q$ (and $F(m^\star)\ge q$), i.e., when $\beta>\beta_0^{\mathrm{IPA}}(m^\star)$; the two conditions differ only at the boundary. For the ratio, the numerators of $\beta_0^\star(m^\star)$ and $\beta_0^{\mathrm{IPA}}(m^\star)$ coincide and are positive for $F(m^\star-1)>q$, so $\beta_0^\star(m^\star)/\beta_0^{\mathrm{IPA}}(m^\star)=F(m^\star-1)/(F(m^\star-1)+c_\varepsilon)$, which is at most $1/(1+c_\varepsilon)=1-e^{-\varepsilon}$ because $F(m^\star-1)\le1$ and tends to it as $F(m^\star-1)\to1$; the asymptotic cost ratio $\beta_0^{\mathrm{IPA}}/\beta_0^\star=1+c_\varepsilon/F(m^\star-1)$ is therefore at least $1/(1-e^{-\varepsilon})$ and below $1+c_\varepsilon/q$. At finite η, both expressions are only sufficient budgets, and by Remark 1 IPA may have no finite certified one. □

H. Proof of Proposition 2

Let $m^\star<x_q$ with $F(m^\star)\le q-\eta$, and let $\beta>0$. Under $s(t)=\mathbf{1}[t\ge m^\star]$, for $t<m^\star$ we have $G_\beta(t)\le(1-\beta)F(m^\star-1)-\beta c_\varepsilon\le q-\eta-\beta c_\varepsilon<q-\eta$. For $t\ge m^\star$, $G_\beta(t)\ge G_\beta(m^\star)=F(m^\star)+\beta(1+c_\varepsilon-F(m^\star))$, which is at least $q+\eta$ if and only if $\beta\ge(q+\eta-F(m^\star))/(1+c_\varepsilon-F(m^\star))$. The curve is non-decreasing (it jumps by $\beta(1+2c_\varepsilon)+(1-\beta)(F(m^\star)-F(m^\star-1))>0$ at $m^\star$), and the accuracy argument of Theorem 1 yields $\hat{m}=m^\star$. Conversely, for any strategy that keeps the curve non-decreasing, the upper envelope of Lemma 1 gives $G_\beta(t)\le F(t)+\beta(1+c_\varepsilon-F(t))$, and the right-hand side is increasing in $F(t)$. If $\beta<(q-\eta-F(m^\star))/(1+c_\varepsilon-F(m^\star))$, then $G_\beta(t)<q-\eta$ for all $t\le m^\star$. Since $m^\star<x_q\le B$, condition (b) gives either $\hat{m}=B>m^\star$ or $G_\beta(\hat{m})>q-\eta$, hence $\hat{m}>m^\star$. For the ratio, write the inflation target as $F(m^\star-1)=q+d$ and the deflation target as $F(m^\star)=q-d$, and set $\eta=0$. The two costs are $d/(q+d+c_\varepsilon)$ and $d/(1-q+d+c_\varepsilon)$, so their ratio is $(q+d+c_\varepsilon)/(1-q+d+c_\varepsilon)$, which tends to $(1+c_\varepsilon)/c_\varepsilon=e^\varepsilon$ as $q\to1$ and $d\to0$. □

I. Proof of Proposition 3

Write $a=p_o-q_o=(e^\varepsilon-1)/(2(e^\varepsilon+1))$. A bit that is 1 with probability s contributes $(s-q_o)/a$ in expectation to the debiased node estimate. With $q_o/a=2/(e^\varepsilon-1)=2c_\varepsilon$ and $p_o/a=(e^\varepsilon+1)/(e^\varepsilon-1)=1+2c_\varepsilon$, the four probabilities 0, p_o+q_o, p_o, and q_o give $-2c_\varepsilon$, $1+2c_\varepsilon$, 1, and 0. These four reporting probabilities lie in $[0,1]$, since $p_o+q_o\le1$. Let $Q=-2c_\varepsilon\,\mathbf{u}_1+(1+2c_\varepsilon)\,\mathbf{u}_{m^\star}$, where $\mathbf{u}_v$ is the unit point mass at v; Q is a signed measure on $[B]$ with $Q([B])=1$. For every node v, the fake contribution is $Q(v)$: it is $-2c_\varepsilon$ if v contains 1 but not $m^\star$, $1+2c_\varepsilon$ if v contains $m^\star$ but not 1, 1 if v contains both, and 0 otherwise. An honest user contributes $P(v)$, the probability mass of v, in expectation. We condition throughout on every level receiving at least one user; under uniform level assignment this fails with probability at most $h(1-1/h)^n$, which can be folded into δ. Condition further on the number $N_\ell\ge1$ of users assigned to level ℓ and the number f_ℓ of fake ones among them. The level-ℓ estimate then has expectation $(1-f_\ell/N_\ell)P(v)+(f_\ell/N_\ell)Q(v)$, and since $\mathbb{E}[f_\ell/N_\ell\mid N_\ell]=\beta$, its unconditional expectation is $(1-\beta)P(v)+\beta Q(v)$. This tree is consistent, because P and Q are measures, and its root is $(1-\beta)+\beta=1$. The post-processing $\mathcal{P}$ is affine and leaves such trees unchanged, so $\mathbb{E}[\mathcal{P}(\text{estimates})]=\mathcal{P}(\mathbb{E}[\text{estimates}])=(1-\beta)P+\beta Q$. Its prefix sums are $(1-\beta)F(t)+\beta Q([1,t])$, where $Q([1,t])=-2c_\varepsilon$ for $1\le t<m^\star$ and $Q([1,t])=1$ for $t\ge m^\star$.

Now let every prefix sum $\hat{S}(t)$ deviate from its expectation by less than η, an event of probability at least $1-\delta$. The empty prefix at $t=0$ is $0<q$. For $1\le t<m^\star$, monotonicity of F and the budget condition give $\hat{S}(t)<(1-\beta)F(m^\star-1)-2\beta c_\varepsilon+\eta\le q$, exactly as in the proof of Theorem 1 with c_ε replaced by $2c_\varepsilon$. If $m^\star<B$, then at $t=m^\star$, $\hat{S}(m^\star)>(1-\beta)F(m^\star)+\beta-\eta\ge F(m^\star)-\eta\ge q$; if $m^\star=B$, the root fixed at 1 gives $\hat{S}(B)=1\ge q$. Hence the first value at which the prefix sum reaches q is $m^\star$. The bound on the budget follows as in Corollary 1: $(F-q+\eta)/(F+2c_\varepsilon)$ is increasing in $F\le1$ (as $\eta<q+2c_\varepsilon$), and $1/(1+2c_\varepsilon)=(e^\varepsilon-1)/(e^\varepsilon+1)=\tanh(\varepsilon/2)$. □

J. Proof of Theorem 3

Fix any strategy with $\beta\le\bar{\beta}$ for which Definition 1 applies to the poisoned curve: one that keeps the curve non-decreasing, or any strategy if Π is robustly accurate. Run Π at level $q^\dagger=q-\bar{\beta}(q+c_\varepsilon)-\eta>0$. Both steps below use only pointwise bounds on G_β.

Upper bound. For $t\ge x_q$, $F(t)\ge q$. By Lemma 1, $G_\beta(t)\ge(1-\beta)q-\beta c_\varepsilon=q-\beta(q+c_\varepsilon)\ge q-\bar{\beta}(q+c_\varepsilon)=q^\dagger+\eta$. With probability at least $1-\delta$, conditions (a) and (b) hold at level $q^\dagger$. By (a), either $\hat{m}=1\le x_q$ or $G_\beta(\hat{m}-1)<q^\dagger+\eta$, so $\hat{m}-1<x_q$; in both cases $\hat{m}\le x_q$.

Lower bound. For t with $F(t)<q^\ddagger$, the upper envelope gives $G_\beta(t)\le(1-\beta)F(t)+\beta(1+c_\varepsilon)\le(1-\bar{\beta})F(t)+\bar{\beta}(1+c_\varepsilon)$, because the middle expression is non-decreasing in β ($1+c_\varepsilon-F(t)>0$). Hence $G_\beta(t)<(1-\bar{\beta})q^\ddagger+\bar{\beta}(1+c_\varepsilon)$, and substituting $q^\ddagger=(q-\bar{\beta}(1+q+2c_\varepsilon)-2\eta)/(1-\bar{\beta})$ gives $(1-\bar{\beta})q^\ddagger+\bar{\beta}(1+c_\varepsilon)=q-\bar{\beta}(q+c_\varepsilon)-2\eta=q^\dagger-\eta$. (If $q^\ddagger\le0$, no $t\ge1$ qualifies and $x_{q^\ddagger}=1$ makes the bound trivial.) By (b), either $\hat{m}=B\ge x_{q^\ddagger}$ or $G_\beta(\hat{m})>q^\dagger-\eta$, in which

case $\hat{m}$ is not among these t. Because F is non-decreasing, these t form a prefix $\{0, \dots, x_{q^\ddagger} - 1\}$, hence $\hat{m} \geq x_{q^\ddagger}$. □

K. Proof of Proposition 4

Let $\mathcal{A}$ be as in the statement, let F be the CDF of a clean population, and set $F'(t) = \min\{1, (F(t)+\bar{\beta}c_\varepsilon)/(1-\bar{\beta})\}$ for $t \in [B]$ and $F'(0) = 0$. F' is non-decreasing with $F'(B) = 1$, so it is the CDF of a population on $[B]$ (with an atom of mass at least $\min\{1, \bar{\beta}c_\varepsilon/(1-\bar{\beta})\}$ at value 1). We assume $\bar{\beta} > 0$; for $\bar{\beta} = 0$ the claim is the hypothesis itself. Define the time-invariant strategy by

$$g(t) = \frac{F(t) - (1-\bar{\beta})F'(t)}{\bar{\beta}} \quad (t \geq 1), \qquad g(0) = 0,$$
$$s(t) = \rho + (1-2\rho)g(t).$$

Where $F'(t) < 1$ we get $g(t) = -c_\varepsilon$. Where $F'(t) = 1$, we have $F(t) \geq 1 - \bar{\beta}(1 + c_\varepsilon)$, so $g(t) = (F(t) - 1 + \bar{\beta})/\bar{\beta} \in [-c_\varepsilon, 1]$. Hence $s(t) \in [0, 1]$, and s is a valid time-invariant OPA strategy. By construction, the poisoned curve of F' under rate $\bar{\beta}$ equals $(1-\bar{\beta})F' + \bar{\beta}g = F$ at every threshold, so it is non-decreasing and s is admissible in the hypothesis. (s itself need not be monotone.) In both worlds, every user's answer to a query at t is an independent Bernoulli bit with mean $\rho + (1-2\rho)F(t)$, whatever thresholds $\mathcal{A}$ chooses. By induction over users, every transcript, and hence every output of $\mathcal{A}$, has the same distribution under "clean population F" and under "population F' poisoned by s at rate $\bar{\beta}$". An estimator that is inflation-proof on F' outputs $\hat{m} \leq x'_q$ with probability at least $1 - \delta$, where $x'_q = \min\{t : F'(t) \geq q\} = \min\{t : F(t) \geq (1-\bar{\beta})q - \bar{\beta}c_\varepsilon\}$. This is the $(q - \bar{\beta}(q + c_\varepsilon))$-quantile of F, and by indistinguishability the same bound holds, with the same probability, on the clean population F. Under IPA, a fake user may pick its input after seeing t, so any g into $[0, 1]$ is feasible; ours is a mixture of steps $\mathbf{1}[t \geq v]$. The analogous construction $F' = \min\{1, F/(1-\bar{\beta})\}$ gives $g = 0$ where $F' < 1$ and $g = (F - 1 + \bar{\beta})/\bar{\beta}$, non-decreasing in $[0, 1]$, where $F' = 1$; it is feasible and yields the shift $\bar{\beta}q$.

Other binary randomizers. Suppose the answer to a query at t is 1 with probability a_t if $x \leq t$ and $b_t < a_t$ otherwise, for any ε-LDP pair (a_t, b_t). Keep F' and g and set $s(t) = b_t + (a_t - b_t)g(t)$. By Lemma 2, $(a_t - b_t)c_\varepsilon \leq b_t$, so $g \geq -c_\varepsilon$ gives $s(t) \geq 0$, and $g \leq 1$ gives $s(t) \leq a_t \leq 1$. An honest user of F' answers 1 with probability $b_t + (a_t - b_t)F'(t)$, so a query at t is answered 1 with probability $b_t + (a_t - b_t)\big((1 - \bar{\beta})F'(t) + \bar{\beta}g(t)\big) = b_t + (a_t - b_t)F(t)$ in the poisoned world, exactly as in the clean world, and the argument above applies unchanged. □

No atom needed. The mass that F' puts at 1 can be spread out. Let $t_0 = x_{q - \bar{\beta}(q + c_\varepsilon)}$ (quantile of F) with $t_0 \geq 2$, and let g_1 be non-increasing with $g_1(1) = 0$ and $g_1(t) = -c_\varepsilon$ for $t \geq t_0$. Set $F' = \min\{1, (F - \bar{\beta}g_1)/(1-\bar{\beta})\}$ (non-decreasing, hence a CDF), and $g = g_1$ where $F' < 1$ and $g = (F - 1 + \bar{\beta})/\bar{\beta} \in [g_1, 1]$ where $F' = 1$. Then $(1-\bar{\beta})F' + \bar{\beta}g = F$ by construction, so the poisoned curve of F' is F; moreover $F'(t) < q$ for $t < t_0$ and $F'(t_0) \geq q$, so the shift is the same. With g_1 decreasing uniformly over $2, \dots, t_0$, every atom of F' is at most (atom of F)$/(1-\bar{\beta}) + \bar{\beta}c_\varepsilon/((1-\bar{\beta})(t_0 - 1))$, so the bound also holds within any class of populations with atoms at most A, for clean populations whose atoms are at most $(1-\bar{\beta})A - \bar{\beta}c_\varepsilon/(t_0 - 1)$.

L. Accuracy of NBS

NBS runs $R = \lceil \log_2 B \rceil$ rounds; round r queries the midpoint t_r of the current interval with a fresh batch of at least $k = \lfloor n/R \rfloor$ users and moves right if the batch estimate $\hat{G}(t_r) < q$ and left otherwise. Fix any curve π', monotone or not. Conditioned on the past, the answers of round r are i.i.d. Bernoulli($\pi'(t_r)$) bits, so by Hoeffding's inequality $|\hat{\pi}(t_r) - \pi'(t_r)| < s$ with probability at least $1 - 2e^{-2ks^2}$, and $|\hat{G}(t_r) - G'(t_r)| < s/(1 - 2\rho_\varepsilon)$. Choosing $s = \sqrt{\ln(2R/\delta)/(2k)}$ and a union bound over the R rounds give $|\hat{G}(t_r) - G'(t_r)| < \eta_{\mathrm{NBS}}$ in every round with probability at least $1 - \delta$. On this event, the final output $\hat{m}$ either equals B or was queried with $\hat{G}(\hat{m}) \geq q$, so $G'(\hat{m}) > q - \eta_{\mathrm{NBS}}$ (condition (b)); and it either equals 1 or was reached after a query at $\hat{m} - 1$ with $\hat{G}(\hat{m} - 1) < q$, so $G'(\hat{m} - 1) < q + \eta_{\mathrm{NBS}}$ (condition (a)). Monotonicity is never used, so NBS is robustly $(\eta_{\mathrm{NBS}}, \delta)$-accurate. Contamination keeps the answers of a round i.i.d., because every contacted user is fake independently with probability β and fake users draw their answers independently. A strategy that chooses its function s_r from the transcript of earlier rounds leaves the answers of round r i.i.d. given that transcript, so the conditional Hoeffding bound and the union bound still apply. Answers that vary within a round are independent but not identical, and Hoeffding's inequality for independent bounded variables applies; answers that depend on earlier answers in the round form a martingale with bounded increments whose conditional means lie in the envelope of Lemma 1, and the Azuma–Hoeffding inequality gives the same bound. □

M. Exact optimality of CLIFF against NBS

Fix n, β, and $m^\star$. NBS is a binary search with a fixed number of rounds, so the output $\hat{m} = m^\star$ determines the query t_r and the required decision of every round: "right" ($\hat{G}(t_r) < q$) if $t_r < m^\star$ and "left" ($\hat{G}(t_r) \geq q$) if $t_r \geq m^\star$; $\hat{G}(t_r)$ is increasing in the number Y_r of "yes" answers of round r. Take any OPA strategy, possibly time-varying, history-dependent, or coordinated among fake users, and couple it with CLIFF on the same batches, fake indicators, and honest randomness. Suppose both have followed the path to $m^\star$ up to round $r-1$; then both query t_r with the same batch and the same honest answers. Under CLIFF, every fake user answers "no" if $t_r < m^\star$ and "yes" otherwise, so $Y_r^{\mathrm{CLIFF}} \leq Y_r$ in the first case and $Y_r^{\mathrm{CLIFF}} \geq Y_r$ in the second, and CLIFF makes the required decision whenever the strategy does. By induction, the event that the strategy outputs $m^\star$ is contained in the event that CLIFF does, so CLIFF maximizes $\Pr[\hat{m} = m^\star]$ at every n and β. □

TABLE VIII
TOY EXAMPLE ($q = 0.9$, $\varepsilon = \ln 3$, $x_q = 8$, $m^\star = 16$, $\beta = 7\%$).

threshold t	8	9	10	12	14	15	16
honest $F(t)$	.900	.940	.970	.990	.998	.999	1.00
margin $F(t) - q$	.000	.040	.070	.090	.098	.099	.100
CLIFF $G_\beta(t)$	.802	.839	.867	.886	.893	**.894**	**1.035**

TABLE IX
VALIDATION OF TABLE II: $\beta_{90}/\hat{\beta}$ / $\beta_{\text{val}}/\hat{\beta}$ (1000 RUNS PER BUDGET); "PL.": PLATEAU.

q	attack	$\varepsilon = 0.25$	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$
0.9	CLIFF	1.00/1.05	1.00/1.05	1.00/1.00	1.00/1.00	1.00/1.05
	IPA	1.00/1.05	1.01/1.05	1.01/1.05	1.00/1.05	1.00/1.05
0.95	CLIFF	1.02/1.05	1.01/1.05	1.00/1.05	1.00/1.00	1.00/1.05
	IPA	1.00/1.05	1.00/1.00	1.01/1.05	1.02/1.05	1.02/1.05
0.99	CLIFF	0.99/1.05	1.01/1.05	1.00/1.05	1.00/1.05	1.00/1.05
	IPA	pl. .838–.898	1.02/1.05	0.99/1.00	1.00/1.05	1.01/1.05
0.999	CLIFF	1.06/1.15	1.08/1.15	1.02/1.15	0.89/0.95	0.99/1.05
	IPA	pl. .30–.32	pl. .43–.44	pl. .63–.64	0.89/1.05	1.00/1.05

APPENDIX B
A TOY EXAMPLE

Consider a domain of $B = 16$ latency buckets with $F(8) = 0.90$, so the p90 is $x_q = 8$, and a thin tail in which F rises from 0.90 to 1 over buckets 9 to 16 (Table VIII). Let $\varepsilon = \ln 3$, so $\rho_\varepsilon = 1/4$ and $c_\varepsilon = 1/2$. The attacker targets the domain maximum, $m^\star = 16 = 2x_q$. Every margin on $[8, 16)$ is at most $1 - q = 0.1$ (Observation 1), and the asymptotic threshold of Theorem 1 is $\beta_0^\star(16) = (0.999 - 0.9)/(0.999 + 0.5) = 6.6\%$. With $\beta = 7\%$, (4) yields $G_\beta(15) = 0.93 \cdot 0.999 - 0.035 = 0.894 < q$ and $G_\beta(16) = 0.93 + 0.105 = 1.035 \geq q$. Binary search probes $8 \to 12 \to 14 \to 15$, finds $G_\beta < q$ at each probe, and returns 16. Without privacy, or under IPA, the same move needs $(0.999-0.9)/0.999 = 9.9\%$ (Proposition 1). Moving the *median* to 16 would need $(0.999 - 0.5)/1.499 = 33\%$.

APPENDIX C
ADDITIONAL RESULTS

Search settings. The tolerances $\hat{\eta}_\pm$ are found by bisection with 11 steps of 200 runs for NBS and 9–10 steps of 60–100 runs for BayeSS and SGD-SN-log (BayeSS still misses $m^\star$ by a few units at larger margins). The empirical budget $\hat{\beta}$ uses 13 steps of 40 trials over $[0,\ 4\max\{\beta_0^\star, \beta_0^{\text{IPA}}\}+1\%]$ (Table II) or $[0, 90\%]$ (RQ4); for Fig. 3(d), the range $[0, 6\%]$ with 13 steps of 40 trials and five bisections for NBS, and 9 steps of 40 trials and three bisections for BayeSS and SGD-SN-log; the planning bisections of RQ3 use $[0, 6\%]$ with 9 steps of 40 trials (the plateau check: 12 steps of 200).

Table IX validates the bisection budgets of Table II. A hit rate close to 0.9 at $\hat{\beta}$ does not by itself locate the budget, since the hit curve can be flat, so we measure the hit curve with 1000 fresh runs at each of 15 budgets around $\hat{\beta}$. For $q \leq 0.99$, $\hat{\beta}$ is within 2.4% of the interpolated 90%-hit budget β_{90} in all 29 measurable cells, and the smallest grid budget with a one-sided 95% Wilson lower bound of 0.9 is at most $1.05\hat{\beta}$. At $q = 0.999$, where the hit curve rises slowly, the deviation is up to 11% and the validated budget up to $1.15\hat{\beta}$. Two kinds of cells have no budget: IPA on the p99.9 for $\varepsilon \leq 1$, whose hit rate stays between 0.30 and 0.64 at every tested budget from 1.4% to 20%, and IPA at $q = 0.99$, $\varepsilon = 0.25$, whose hit rate plateaus at 0.838–0.898 at the tested budgets from 2% to 40%; there the bisection's $2.51 \pm 0.64\%$ reflects a plateau near the criterion, as the prediction ∞ of Remark 1 anticipates. The standard deviation over five bisections describes the spread of the search, not the accuracy of $\hat{\beta}$, which this validation measures.

TABLE X
RQ3: ATTACKER KNOWLEDGE. CLIFF USES $1.2\tilde{\beta}$, MGA $\tilde{\beta}$, WITH $\tilde{\beta}$ (%) ESTIMATED FROM N_{aux} SAMPLES.

		NBS				BayeSS			
		MGA		CLIFF		MGA		CLIFF	
N_{aux}	$\tilde{\beta}$ (%)	hit↑	log-err↓	hit↑	log-err↓	hit↑	log-err↓	hit↑	log-err↓
10^3	0.63 ± 0.01	0.05 ± 0.06	1.36 ± 0.71	0.60 ± 0.22	0.46 ± 0.62	0.15 ± 0.12	1.26 ± 0.75	0.55 ± 0.19	0.53 ± 0.74
10^4	0.63 ± 0.00	0.08 ± 0.10	1.35 ± 0.75	0.60 ± 0.09	0.41 ± 0.59	0.00 ± 0.00	1.18 ± 0.69	0.40 ± 0.25	0.42 ± 0.46
10^5	0.63 ± 0.00	0.05 ± 0.06	1.36 ± 0.61	0.53 ± 0.18	0.40 ± 0.52	0.10 ± 0.20	1.31 ± 0.76	0.50 ± 0.16	0.39 ± 0.57
10^6	0.63 ± 0.00	0.00 ± 0.00	1.54 ± 0.47	0.48 ± 0.05	0.43 ± 0.51	0.05 ± 0.10	1.03 ± 0.60	0.45 ± 0.24	0.44 ± 0.49

Table X complements RQ3. The attacker estimates the budget of Theorem 1 from an auxiliary sample of size N_{aux} drawn from the same population and then runs MGA with exactly the estimated budget $\tilde{\beta}$, or CLIFF with $1.2\tilde{\beta}$ (CreditCard, $\kappa = 10$; each trial draws its own auxiliary sample, so $\pm$ includes the noise of $\tilde{\beta}$). Even $N_{\text{aux}} = 10^3$ estimates $\beta^\star$ almost exactly, but $1.2\tilde{\beta}$ does not cover the protocol error at $n = 10^6$, so CLIFF hits the target in only 40–60% of runs, with a landing error below one octave. MGA hits it in at most 15% of runs, with a landing error above one octave, because without a step its landing point depends on the exact budget. Planning the budget against the public protocol on a synthetic population accounts for the protocol error and succeeds (Table IV); so does over-provisioning CLIFF to two or three times the universal bound (Fig. 4).

Table XI separates two-sided from one-sided success on the RQ3 grid. MGA reaches the target at least as often as CLIFF but, under NBS and BayeSS from $1.5\beta^\star$ on, mostly lands in the top half of the domain; CLIFF reaches the target exactly when it hits it, and S-CLIFF nearly so. Once its search passes $m^\star$, MGA never again queries just below $m^\star$, where the margin is thin; under SGD-SN-log, its push toward B also hides the averaging lag (reach 0.90 at $1.5\beta^\star$, edge 0.83 at $2\beta^\star$). Even random answers inflate the p99 ($24 \pm 14\times$ at $3\beta^\star$, NBS); the step makes the attack precise. Table XII repeats the planning of Table IV when the attacker's guess of x_q is wrong.

Without HH's consistency step (CreditCard, $n = 10^7$, 40 runs), clean HH is erratic (rank 0.93±0.06) and generic CLIFF at 1% sends the output to the domain edge ($51.5 \times x_q$) in every run; with it, the output stays at $1.14 \times x_q$, and generic CLIFF hits the target in 0.58 of runs at 2% (separate runs from Table III) and 1.00 at 3%. With a fixed $\eta = 0.001$

TABLE XI
RQ3: TWO-SIDED (HIT) VERSUS ONE-SIDED (REACH) SUCCESS, AND ESTIMATES IN THE TOP HALF OF THE DOMAIN (EDGE).

		1×			1.5×			2×			3×		
	attack	hit	reach	edge	hit	reach	edge	hit	reach	edge	hit	reach	edge
NBS	CLIFF	0.30	0.30	0.00	0.88	0.88	0.00	0.98	0.98	0.00	1.00	1.00	0.00
	S-CLIFF	0.23	0.28	0.05	0.80	0.80	0.00	0.95	0.98	0.00	1.00	1.00	0.00
	MGA	0.03	0.80	0.45	0.00	0.98	0.85	0.00	1.00	1.00	0.00	1.00	1.00
	IPA	0.10	0.15	0.05	0.40	0.40	0.00	0.60	0.63	0.00	1.00	1.00	0.00
	FGA-R	0.38	0.38	0.00	0.35	0.38	0.00	0.30	0.35	0.03	0.40	0.40	0.00
BayeSS	CLIFF	0.45	0.45	0.00	0.90	0.90	0.00	1.00	1.00	0.00	1.00	1.00	0.00
	S-CLIFF	0.45	0.45	0.00	0.85	0.85	0.00	1.00	1.00	0.00	0.95	1.00	0.00
	MGA	0.10	0.90	0.40	0.00	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00
	IPA	0.05	0.05	0.00	0.20	0.20	0.00	0.70	0.70	0.00	0.90	0.95	0.00
	FGA-R	0.25	0.25	0.00	0.20	0.20	0.00	0.15	0.15	0.00	0.30	0.30	0.00
SGD-SN-log	CLIFF	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.08	0.00	0.93	0.93	0.00
	S-CLIFF	0.00	0.00	0.00	0.00	0.00	0.00	0.13	0.13	0.00	0.98	0.98	0.00
	MGA	0.00	0.00	0.00	0.08	0.90	0.08	0.00	1.00	0.83	0.00	1.00	1.00
	IPA	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.10	0.00
	FGA-R	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

instead of the margins of Table VII ($\beta = \bar{\beta}$, same nine settings per protocol), CTQ kept the output at or below x_q in 0.35–0.80 of runs for NBS, 0.05–0.75 for BayeSS, and 0.68–1.00 for SGD-SN-log (RQ5). Table XVII gives the HH audit of RQ5 ($\varepsilon = 1$, $\kappa = 10$; 200 runs, 100 at 10^9; a run is flagged if some raw node estimate lies below $-z^\star$ standard errors, with $z^\star$ the Bonferroni level at 1% over all nodes; the spread variant removes the mass over $[1, x_q/2]$). Table XIII gives RQ6(a) (the attacker's 5,000 transactions are log-uniform in $(x_q, 10x_q]$; CTQ uses $\bar{\beta} = 1\%$ and the margins of RQ5) and Table XV RQ6(b) ($n = 10^6$; inflation $2^{\bar{\theta}_n}/x_q$ of the averaged $\log_2$-scale iterate; width of the 95% interval in $\log_2$ units). Table XVI gives the continuous-data control of RQ6(b) (each CreditCard bucket spread uniformly; clean: 1000 runs, Wilson intervals; attacked: 200 runs; raw-scale iterates on values in units of the true p99; Table XV uses 100 runs per cell and a looser criterion, an interval that meets the bucket of x_q, so its lower clean coverage reflects sampling noise); Table XIV the false SLO alarms of RQ6 (an alarm fires when the estimated p99 delay exceeds twice the true one, 274 min; CLIFF targets three times, 411 min), which fire in every run at 1% (clean: at most 5%); Table XVIII the direction asymmetry of RQ4 (success: $\hat{m} \geq x_{q+d}$, resp. $\hat{m} \leq x_{q-d}$, in $\geq 90\%$ of runs; "—": $q + d > 1$; for $q = 0.99$ and $d = 0.01$ the inflation target is the largest data value); and Fig. 5 the tail-shape study of RQ4. In Table XVI, the original raw-scale iterate, on values in units of the true p99 (an oracle scaling for this control) with the default step size, under-covers clean data (0.883); under attack it covers the true p99 in at most 0.05 of runs from 0.5% on, and $m^\star$ in 0.69 of runs at 1% and in all from 2%. At 1%, the log-scale interval covers $m^\star$ partly by widening (5.6× the clean width), and at 5% it is narrower than clean.

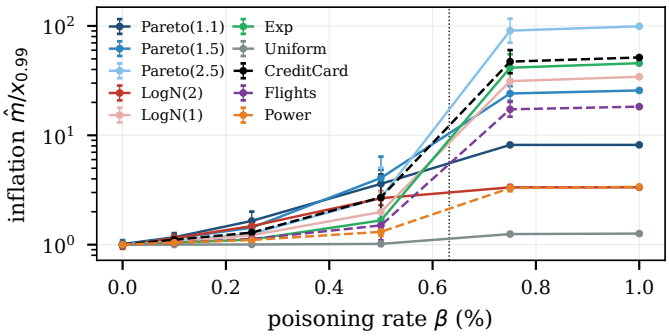


Fig. 5. RQ4: inflation of the p99 achievable at rate β (CLIFF at B, NBS, $n = 10^7$, $\varepsilon = 1$; geometric mean over 40 runs, bars: one standard deviation of the log-inflation). The dotted line marks $(1 - q)(1 - e^{-\varepsilon})$.

TABLE XII
RQ3: PLANNING WITH GUESSED INFLATION $\hat{\kappa}$: PLAN (%) / HIT RATE WITH THE 25% MARGIN.

	CreditCard				Flights				Power			
protocol (n)	$\hat{\kappa}$=2	5	10	1000	2	5	10	1000	2	5	10	1000
NBS (10^6)	0.73/0.83	0.96/0.95	1.14/1.00	1.05/1.00	0.67/0.88	0.86/1.00	0.86/1.00	0.86/1.00	0.74/0.80	1.00/1.00	1.15/1.00	1.01/1.00
BayeSS (10^6)	0.66/0.73	0.93/1.00	0.97/1.00	1.02/1.00	0.62/0.68	0.88/1.00	0.97/1.00	0.98/1.00	0.74/0.85	1.00/1.00	1.00/1.00	1.03/1.00
SGD-SN-log (10^6)	0.70/0.00	1.24/0.45	1.80/1.00	3.75/1.00	0.70/0.00	1.24/0.90	1.71/1.00	2.47/1.00	0.76/0.58	1.24/1.00	1.80/1.00	3.38/1.00
SGD-SN-log (10^7)	0.46/0.00	0.70/1.00	0.77/1.00	1.42/1.00	0.45/0.00	0.70/0.95	0.77/0.98	1.14/1.00	0.46/0.00	0.70/1.00	0.77/1.00	1.29/1.00
HH (T-CLIFF) (10^7)	1.14/1.00	1.28/1.00	1.27/1.00	1.29/1.00	0.63/0.55	0.83/1.00	0.89/1.00	0.90/1.00	0.75/0.60	0.97/1.00	0.98/1.00	1.03/1.00

TABLE XIII
RQ6(A): ATTACKER TRANSACTIONS THAT EVADE REVIEW (%) AND REVIEW THRESHOLD (EUR).

		evading review (%)↑				threshold (EUR)	
		$\beta = 0$	0.5%	1%	2%	clean	$\beta = 1\%$
NBS	plain	8 ± 21	57 ± 27	96 ± 13	100 ± 0	1530 ± 2090	9511 ± 1876
	CTQ	0 ± 0	0 ± 0	0 ± 0	4 ± 9	410 ± 17	558 ± 39
BayeSS	plain	7 ± 9	67 ± 27	100 ± 1	100 ± 0	1208 ± 314	10210 ± 393
	CTQ	0 ± 0	0 ± 0	0 ± 0	21 ± 19	452 ± 20	645 ± 48
SGD-SN-log	plain	1 ± 2	30 ± 9	87 ± 3	97 ± 1	1035 ± 64	7641 ± 529
	CTQ	0 ± 0	0 ± 0	0 ± 0	73 ± 11	533 ± 14	850 ± 39

TABLE XIV
RQ6: RATE OF FALSE SLO ALARMS ON FLIGHTS.

	alarm rate↑					
protocol	$\beta = 0$	0.25%	0.5%	0.75%	1%	2%
NBS	0.00 ± 0.00	0.03 ± 0.05	0.33 ± 0.17	0.88 ± 0.14	1.00 ± 0.00	1.00 ± 0.00
BayeSS	0.05 ± 0.10	0.10 ± 0.12	0.45 ± 0.19	0.95 ± 0.10	1.00 ± 0.00	1.00 ± 0.00
SGD-SN-log	0.00 ± 0.00	0.00 ± 0.00	0.13 ± 0.14	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00

TABLE XV
RQ6(B): CONFIDENCE HIJACKING OF SGD-SN-LOG (100 RUNS).

data	β	inflation	CI width	cover x_q↓	cover $m^\star$↑
CreditCard	0%	1.01 ± 0.06	0.376 ± 0.180	0.89 ± 0.06	0.00 ± 0.00
	0.5%	1.97 ± 0.31	0.991 ± 0.501	0.04 ± 0.04	0.00 ± 0.00
	1%	7.63 ± 0.73	2.185 ± 0.654	0.00 ± 0.00	1.00 ± 0.00
	2%	9.31 ± 0.15	0.783 ± 0.171	0.00 ± 0.00	1.00 ± 0.00
	5%	9.77 ± 0.04	0.298 ± 0.040	0.00 ± 0.00	1.00 ± 0.00
Flights	0%	1.02 ± 0.02	0.160 ± 0.071	0.91 ± 0.09	0.00 ± 0.00
	0.5%	1.51 ± 0.22	0.563 ± 0.390	0.03 ± 0.02	0.00 ± 0.00
	1%	7.86 ± 0.76	1.949 ± 0.747	0.00 ± 0.00	1.00 ± 0.00
	2%	9.54 ± 0.12	0.520 ± 0.136	0.00 ± 0.00	1.00 ± 0.00
	5%	9.91 ± 0.02	0.144 ± 0.025	0.00 ± 0.00	1.00 ± 0.00

TABLE XVI
RQ6(B) CONTROL: COVERAGE BY SGD-SN WITH LOG- AND RAW-SCALE ITERATES.

β	log, discrete	log, continuous	raw, continuous
clean	0.932 [0.91, 0.95]	0.945 [0.93, 0.96]	0.883 [0.86, 0.90]
0.5%	–	0.04 / 0.00 / 2.1 / 2.8	0.02 / 0.00 / 2.1 / 4.0
1%	–	0.00 / 1.00 / 7.7 / 5.6	0.05 / 0.69 / 6.4 / 33.3
2%	–	0.00 / 1.00 / 9.3 / 2.1	0.00 / 1.00 / 9.4 / 17.2
5%	–	0.00 / 1.00 / 9.8 / 0.8	0.00 / 1.00 / 9.9 / 4.1

TABLE XVII
HH NON-NEGATIVITY AUDIT: HIT RATE / FRACTION OF RUNS FLAGGED.

dataset	n	β	T-CLIFF	spread	clean
CreditCard	10^7	0%	—	—	0.00 / 0.020
	10^7	1%	0.06 / 0.030	0.07 / 0.005	
	10^7	1.5%	1.00 / 0.130	1.00 / 0.015	
	10^8	0%	—	—	0.00 / 0.010
	10^8	0.75%	0.99 / 0.015	0.99 / 0.015	
Flights	10^7	0%	—	—	0.00 / 0.020
	10^7	1%	1.00 / 1.000	1.00 / 0.015	
	10^7	1.5%	1.00 / 1.000	1.00 / 0.010	
	10^8	0%	—	—	0.00 / 0.010
	10^8	0.75%	1.00 / 1.000	1.00 / 0.050	
Power	10^7	0%	—	—	0.00 / 0.010
	10^7	1%	0.92 / 1.000	0.92 / 0.005	
	10^7	1.5%	1.00 / 1.000	1.00 / 0.005	
	10^8	0%	—	—	0.00 / 0.010
	10^8	0.75%	1.00 / 1.000	1.00 / 0.010	
Pareto(1.5)	10^7	0%	—	—	0.00 / 0.015
	10^7	1%	0.03 / 0.685	0.03 / 0.005	
	10^7	1.5%	1.00 / 1.000	1.00 / 0.015	
	10^8	0%	—	—	0.00 / 0.020
	10^8	0.75%	1.00 / 1.000	1.00 / 0.010	
Flights	10^8	1.5%	—	1.00 / 0.900	
Flights	10^9	0.75%	—	1.00 / 1.000	

TABLE XVIII
RQ4: BUDGET (%) TO MOVE THE q-TH QUANTILE UP TO x_{q+d} OR DOWN TO x_{q-d}, AND THE DOWN/UP RATIO (PROPOSITION 2).

		inflate (up)		deflate (down)		ratio	
q	d	emp.↓	thm.	emp.↓	thm.	emp.	thm.
0.95	0.002	0.29 ± 0.01	0.13	0.76 ± 0.02	0.31	2.65	2.40
0.95	0.005	0.47 ± 0.01	0.32	1.14 ± 0.05	0.78	2.41	2.42
0.95	0.01	0.80 ± 0.01	0.65	1.95 ± 0.03	1.55	2.43	2.39
0.95	0.02	1.46 ± 0.01	1.29	3.51 ± 0.02	3.05	2.40	2.37
0.95	0.04	2.71 ± 0.01	2.54	6.32 ± 0.02	5.93	2.33	2.33
0.99	0.002	0.31 ± 0.02	0.13	0.83 ± 0.02	0.34	2.67	2.66
0.99	0.005	0.49 ± 0.01	0.32	1.31 ± 0.03	0.84	2.64	2.64
0.99	0.01	0.85 ± 0.01	0.63	2.04 ± 0.06	1.66	2.40	2.63
0.99	0.02	—		3.69 ± 0.02	3.27	—	—
0.99	0.04	—		6.77 ± 0.03	6.33	—	—